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# MODULE II

| II | Power Transmission<br>Transmission Line Parameters: Resistance, inductance and capacitance of 1- $\phi$ , 2 wire lines-composite conductors (Derivation Required).                                                                                                                                                                                                                                                                                             | 10 | 15% |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
|    | Inductance and capacitance of 3- $\phi$ lines. Symmetrical and unsymmetrical spacing-transposition-double circuit lines-bundled conductors (Derivation Required) .Numerical Problems<br><br><b>Modelling of Transmission Lines:</b> Classification of lines-short lines-voltage regulation and efficiency-medium lines-nominal T and $\Pi$ configurations-ABCD constants- long lines- rigorous solution- interpretation of long line equation-Ferranti effect. |    |     |



# Constants of Transmission line

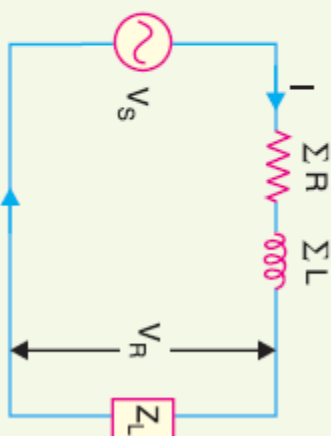
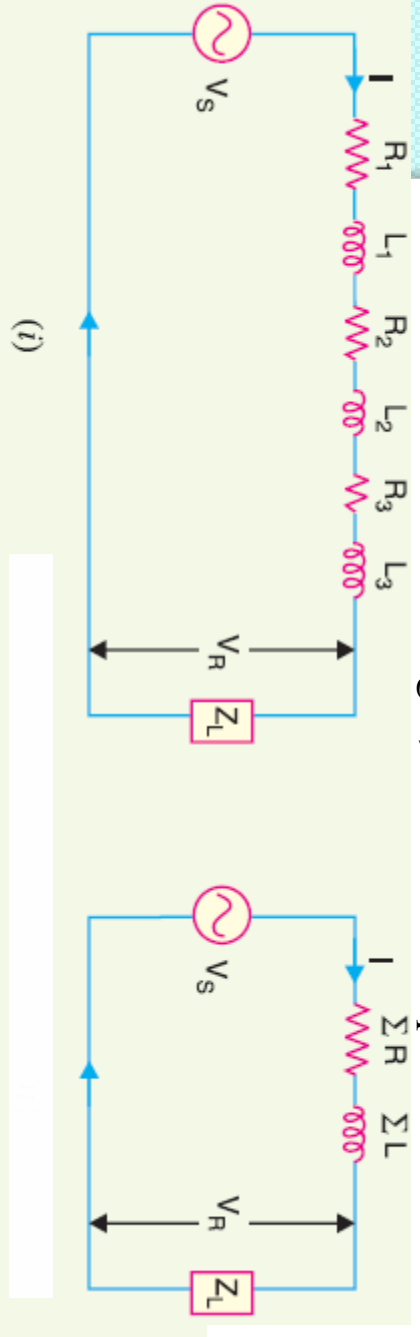
- A transmission line has resistance, inductance and capacitance uniformly distributed along the whole length of the line.

## 1. Resistance

- It is the opposition of line conductors to current flow.
- The resistance is distributed uniformly along the whole length of the line as shown in Fig (i).
- For analysis we convert distributed to lumped parameter as shown in fig(ii).

## 2. Inductance

- When an alternating current flows through a conductor, a changing flux is set up which links the conductor.
- The inductance is also uniformly distributed along the length of the line as shown in Fig. (i).
- For analysis we convert distributed to lumped parameter as shown in fig(ii).
- Due to these flux linkages, the conductor possesses inductance.



$$L = \frac{\Psi}{I} \text{ henry}$$

$\Psi$  = flux linkages in weber-turns

$I$  = current in amperes

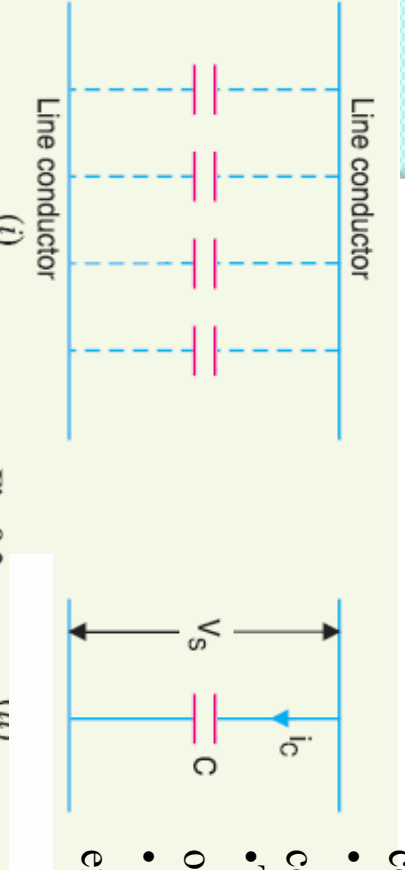


### 3. Capacitance

- We know that any two conductors separated by an insulating material constitute a capacitor.
- As any two conductors of an overhead transmission line are separated by air which acts as an insulation, therefore, capacitance exists between any two overhead line conductors.
- The capacitance between the conductors is the charge per unit potential difference i.e.,

$$\text{Capacitance, } C = \frac{q}{V} \text{ farad}$$

- The capacitance is uniformly distributed along the whole length of the line and may be regarded as a uniform series of capacitors connected between the conductors as shown in Fig. (i).



- When an alternating voltage is impressed on a transmission line, there is a change in the charge on the conductors.
- The result is that a charging current flowing between the conductors.
- This charging current flows in the line even when it is open-circuited i.e., supplying no load.
- It affects the voltage drop along the line as well as the efficiency and power factor of the line.



# Resistance

- The resistance of transmission line conductors is the most important cause of power loss in a transmission line.
- The resistance  $R$  of a line conductor having resistivity  $r$ , length  $l$  and area of cross-section  $a$  is given by

$$R = \rho \frac{l}{a}$$

- The variation of resistance of metallic conductors with temperature is practically linear over the normal range of operation. Suppose  $R_1$  and  $R_2$  are the resistances of a conductor at  $t_1$  °C and  $t_2$  °C.

- ( $t_1 > t_2$ ) respectively If  $\alpha_1$  is the temperature coefficient at  $t_1$  °C, then,

$$\text{where } \alpha_1 = \frac{\alpha_0}{1 + \alpha_0 t_1}$$

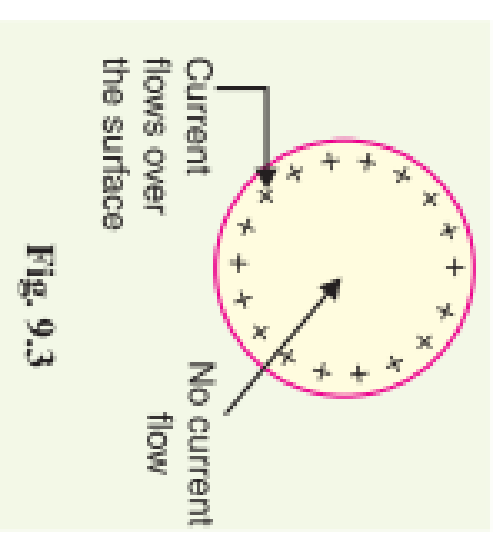
$\alpha_0$  = temperature coefficient at 0° C

- In a single phase or 2-wire d.c line, the total resistance (known as *loop resistance*) is equal to double the resistance of either conductor.
- In case of a 3-phase transmission line, resistance per phase is the resistance of one conductor.



# Skin Effect

- The tendency of alternating current to concentrate near the surface of a conductor is known as **skin effect**.
- When a conductor is carrying steady direct current (d.c.), this current is uniformly distributed over the whole X-section of the conductor.
- However, an alternating current flowing through the conductor does not distribute uniformly, rather it has the tendency to concentrate near the surface of the conductor as shown in Fig. This is known as skin effect.
- Due to skin effect, the effective area of cross-section of the conductor through which current flows is reduced.
- Consequently, the resistance of the conductor is slightly increased when carrying an alternating current.



# Causes of skin effect

- A solid conductor may be thought to be consisting of a large number of strands, each carrying a small part of the current.
- The inductance of each strand will vary according to its position.
- Thus, the strands near the centre are surrounded by a greater magnetic flux and hence have larger inductance than that near the surface.
- The high reactance of inner strands causes the alternating current to flow near the surface of conductor. This crowding of current near the conductor surface is the skin effect.
- The skin effect depends upon the following factors :
  - Nature of material
  - Diameter of wire
    - increases with the diameter of wire.
  - Frequency
    - increases with the increase in frequency.
  - Shape of wire
    - less for stranded conductor than the solid conductor.
- It may be noted that skin effect is negligible when the supply frequency is low ( $< 50$  Hz) and conductor diameter is small ( $< 1$  cm).

# Inductance

- We know that inductance

$$L = \frac{\psi}{I} \text{ henry}$$

$\psi$  = flux linkages in weber-turns

$I$  = current in amperes

- To find L value we have to first find out the flux linkages.

## Flux linkages

### 1. Flux linkages due to a single current carrying conductor

- Consider a long straight cylindrical conductor of radius  $r$  metres and carrying a current  $I$  amperes (r.m.s.) as shown in Fig.
- This current will set up magnetic field.
- The magnetic lines of force will exist inside the conductor as well as outside the conductor.
- Both these fluxes will contribute to the inductance of the conductor.

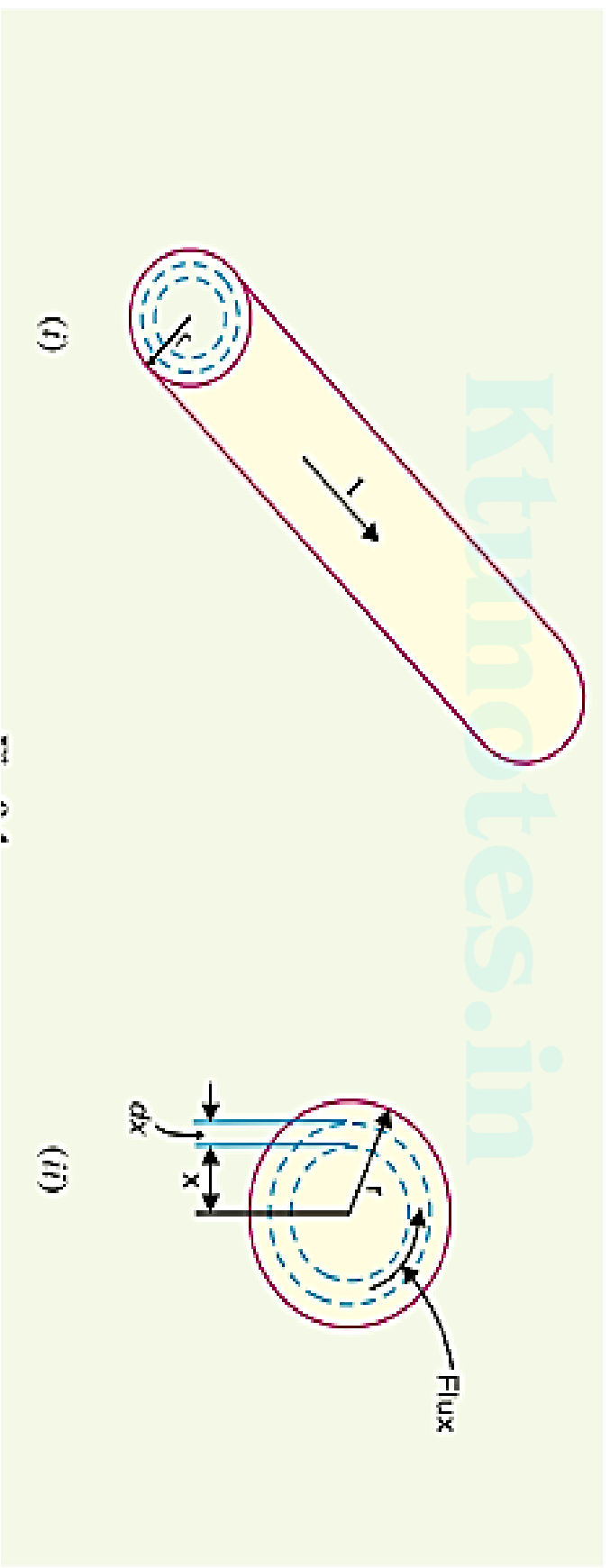
(i) **Flux linkages due to internal flux.** Refer to Fig. 9.4 (ii) where the X-section of the conductor is shown magnified for clarity. The magnetic field intensity at a point  $x$  metres from the centre is given by;

$$*H_x = \frac{I_x}{2\pi x}$$

Assuming a uniform current density,

$$I_x = \frac{\pi x^2}{\pi r^2} I = \frac{x^2}{r^2} I$$

$$\therefore H_x = \frac{x^2}{r^2} \times I \times \frac{1}{2\pi x} = \frac{x}{2\pi r^2} I \text{ AT/m}$$



If  $\mu$  ( $= \mu_0 \mu_r$ ) is the permeability of the conductor, then flux density at the considered point is given by;

$$\begin{aligned} B_x &= \mu_0 \mu_r H_x \text{ wb/m}^2 \\ &= \frac{\mu_0 \mu_r x}{2 \pi r^2} I = \frac{\mu_0 x I}{2 \pi r^2} \text{ wb/m}^2 \quad [\because \mu_r = 1 \text{ for non-magnetic material}] \end{aligned}$$

Now, flux  $d\phi$  through a cylindrical shell of radial thickness  $dx$  and axial length 1 m is given by;

$$d\phi = B_x \times 1 \times dx = \frac{\mu_0 x I}{2 \pi r^2} dx \text{ weber}$$

This flux links with current  $I_x \left( = \frac{I \pi x^2}{\pi r^2} \right)$  only. Therefore, flux linkages per metre length of the conductor is

$$d\psi = \frac{\pi x^2}{\pi r^2} d\phi = \frac{\mu_0 I x^3}{2 \pi r^4} dx \text{ weber-turns}$$

Total flux linkages from centre upto the conductor surface is

$$\begin{aligned} \psi_{int} &= \int_0^r \frac{\mu_0 I x^3}{2 \pi r^4} dx \\ &= \frac{\mu_0 I}{8 \pi} \text{ weber-turns per metre length} \end{aligned}$$

(ii) **Flux linkages due to external flux.** Now let us calculate the flux linkages of the conductor due to external flux. The external flux extends from the surface of the conductor to infinity. Referring to Fig. 9.5, the field intensity at a distance  $x$  metres (from centre) outside the conductor is given by ;

$$H_x = \frac{I}{2 \pi x} \text{ AT / m}$$

$$\text{Flux density, } B_x = \mu_0 H_x = \frac{\mu_0 I}{2 \pi x} \text{ wb/m}^2$$

Now, flux  $d\phi$  through a cylindrical shell of thickness  $dx$  and axial length  $l$  metre is

$$d\phi = B_x dx = \frac{\mu_0 I}{2 \pi x} dx \text{ webers}$$

The flux  $d\phi$  links all the current in the conductor once and only once.

$$\therefore \text{ Flux linkages, } d\psi = d\phi = \frac{\mu_0 I}{2 \pi x} dx \text{ weber-turns}$$

Total flux linkages of the conductor from surface to infinity,

$$\psi_{\text{ext}} = \int_r^{\infty} \frac{\mu_0 I}{2 \pi x} dx \text{ weber-turns}$$

$$\therefore \text{ Overall flux linkages, } \psi = \psi_{\text{int}} + \psi_{\text{ext}} = \frac{\mu_0 I}{8 \pi} + \int_r^{\infty} \frac{\mu_0 I}{2 \pi x} dx$$

$$\therefore \psi = \frac{\mu_0 I}{2 \pi} \left[ \frac{1}{4} + \int_r^{\infty} \frac{dx}{x} \right] \text{ wb-turns/m length}$$

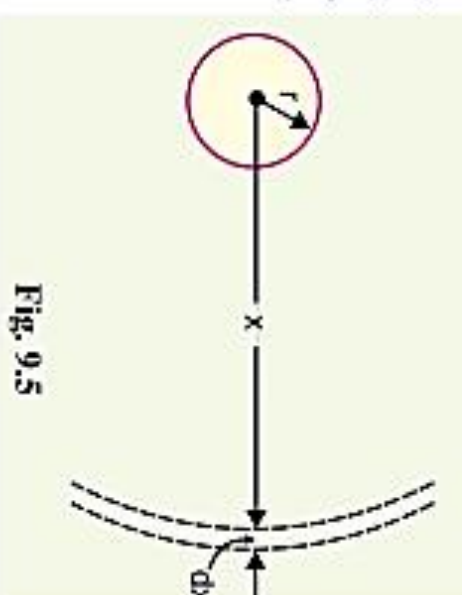


Fig. 9.5

## 2. Flux linkages in parallel current-carrying conductors.

We shall now determine the flux linkages in a group of parallel current carrying conductors. Fig. 9.6 shows the conductors  $A, B, C$  etc. carrying currents  $I_A, I_B, I_C$  etc. Let us consider the flux linkages with one conductor, say conductor  $A$ . There will be flux linkages with conductor  $A$  due to its own current as discussed previously. Also there will be flux linkages with this conductor due to the mutual inductance effects of  $I_B, I_C, I_D$  etc. We shall now determine the total flux linkages with conductor  $A$ .

Flux linkages with conductor  $A$  due to its own current

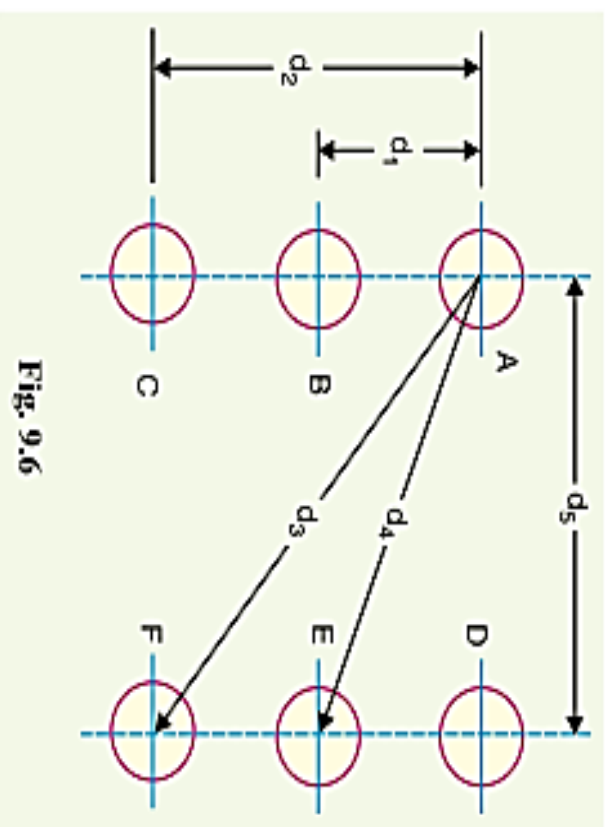


Fig. 9.6

$$\begin{aligned} &= \frac{\mu_0 I_A}{2\pi} \left[ \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right] \quad \dots(i) \\ \text{Flux linkages with conductor } A \text{ due to current } I_B \end{aligned}$$

$$\begin{aligned} &= \frac{\mu_0 I_B}{2\pi} \int_{d_1}^\infty \frac{dx}{x} \quad \dots(ii) \end{aligned}$$

Flux linkages with conductor  $A$  due to current  $I_C$

$$\begin{aligned} &= \frac{\mu_0 I_C}{2\pi} \int_{d_1}^\infty \frac{dx}{x} \quad \dots(iii) \end{aligned}$$

$$\therefore \text{Total flux linkages with conductor } A = (i) + (ii) + (iii) + \dots$$

$$\begin{aligned} &= \frac{\mu_0 I_A}{2\pi} \left( \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right) + \frac{\mu_0 I_B}{2\pi} \int_{d_1}^\infty \frac{dx}{x} + \frac{\mu_0 I_C}{2\pi} \int_{d_2}^\infty \frac{dx}{x} + \dots \end{aligned}$$

Similarly, flux linkages with other conductors can be determined. The above relation provides the basis for evaluating inductance of any circuit.

# Inductance of a single phase two wire line

Consider a single phase overhead line consisting of two parallel conductors  $A$  and  $B$  spaced  $d$  metres apart as shown in Fig. 9.7. Conductors  $A$  and  $B$  carry the same amount of current (*i.e.*  $I_A = I_B$ ), but in the opposite direction because one forms the return circuit of the other.

$$\therefore I_A + I_B = 0$$

In order to find the inductance of conductor  $A$  (or conductor  $B$ ), we shall have to consider the flux linkages with it. There will be flux linkages with conductor  $A$  due to its own current  $I_A$  and also due to the mutual inductance effect of current  $I_B$  in the conductor  $B$ .

Flux linkages with conductor  $A$  due to its own current

$$= \frac{\mu_0 I_A}{2\pi} \left( \frac{1}{4} + \int_r^{\infty} \frac{dx}{x} \right) \quad \dots(i) \quad [\text{See Art. 9.4}]$$

Flux linkages with conductor  $A$  due to current  $I_B$

$$= \frac{\mu_0 I_B}{2\pi} \int_d^{\infty} \frac{dx}{x} \quad \dots(ii)$$

Total flux linkages with conductor  $A$  is

$$W = \text{exn. (i)} + \text{exn. (ii)}$$

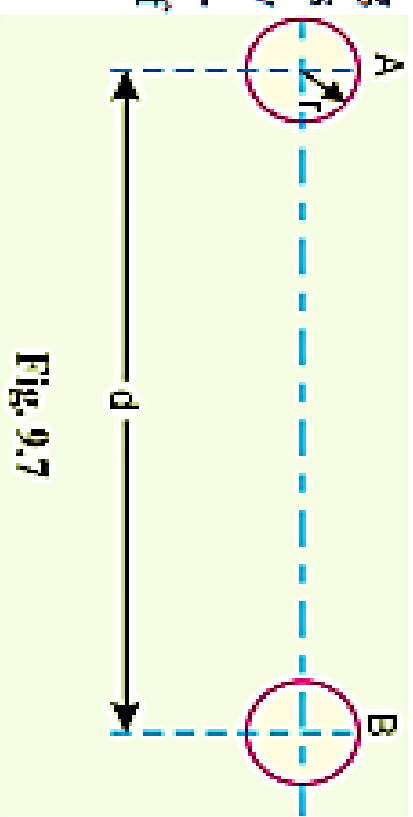


Fig. 9.7

Total flux linkages with conductor A is

$$\Psi_A = \exp. (i) + \exp (ii)$$

$$= \frac{\mu_0 I_A}{2\pi} \left( \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right) + \frac{\mu_0 I_B}{2\pi} \int_d^\infty \frac{dx}{x}$$

$$= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right) I_A + I_B \int_d^\infty \frac{dx}{x} \right]$$

$$= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} + \log_e \infty - \log_e r \right) I_A + (\log_e \infty - \log_e d) I_B \right]$$

$$= \frac{\mu_0}{2\pi} \left[ \left( \frac{I_A}{4} + \log_e \infty (I_A + I_B) - I_A \log_e r - I_B \log_e d \right) \right]$$

$$= \frac{\mu_0}{2\pi} \left[ \frac{I_A}{4} - I_A \log_e r - I_B \log_e d \right] \quad (\because I_A + I_B = 0)$$

Now,

$$\therefore \quad I_A + I_B = 0 \quad \text{or} \quad -I_B = I_A$$

$$-I_B \log_e d = I_A \log_e d$$

$$\therefore \quad \Psi_A = \frac{\mu_0}{2\pi} \left[ \frac{I_A}{4} + I_A \log_e d - I_A \log_e r \right] \text{ wb-turns/m}$$

$$= \frac{\mu_0}{2\pi} \left[ \frac{I_A}{4} + I_A \log_e \frac{d}{r} \right]$$

$$= \frac{\mu_0 I_A}{2\pi} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right] \text{ wb-turns/m}$$

$$\text{Inductance of conductor A, } L_A = \frac{\Psi_A}{I_A}$$

$$= \frac{\mu_0}{2\pi} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right] \text{ H/m} = \frac{4\pi \times 10^{-7}}{2\pi} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right] \text{ H/m}$$

$$\therefore L_A = 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{d}{r} \right] \text{ H / m} \quad \dots(i)$$

$$\text{Loop inductance} = 2 L_A \text{ H/m} = 10^{-7} \left[ 1 + 4 \log_e \frac{d}{r} \right] \text{ H / m}$$

$$\therefore \text{Loop inductance} = 10^{-7} \left[ 1 + 4 \log_e \frac{d}{r} \right] \text{ H / m} \quad \dots(ii)$$

Note that eq. (ii) is the inductance of the two-wire line and is sometimes called loop inductance. However, inductance given by eq. (i) is the inductance per conductor and is equal to half the loop inductance.

**Expression in alternate form.** The expression for the inductance of a conductor can be put in a concise form.

$$L_A = 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{d}{r} \right] \text{ H / m}$$

$$= 2 \times 10^{-7} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right]$$

$$\therefore L_A = 2 \times 10^{-7} \left[ \log_e e^{1/4} + \log_e \frac{d}{r} \right]$$

If we put  $r e^{-1/4} = r'$ , then,

$$L_A = 2 \times 10^{-7} \log_e \frac{d}{r'} \text{ H / m} \quad \dots(iii)$$

The radius  $r'$  is that of a fictitious conductor assumed to have no internal flux but with the same inductance as the actual conductor of radius  $r$ . The quantity  $e^{-1/4} = 0.7788$  so that

$$r' = r e^{-1/4} = 0.7788 r$$

The term  $r'$  ( $= r e^{-1/4}$ ) is called **geometric mean radius (GMR)** of the wire. Note that eq. (iii) gives the same value of inductance  $L_A$  as eq. (i). The difference is that eq. (iii) omits the term to account for internal flux but compensates for it by using an adjusted value of the radius of the conductor.

$$\text{Loop inductance} = 2 L_A = 2 \times 2 \times 10^{-7} \log_e \frac{d}{r'} \text{ H/m}$$

Note that  $r' = 0.778$

# Inductance of 3phase overhead line

Fig. 9.8 shows the three conductors  $A$ ,  $B$  and  $C$  of a 3-phase line carrying currents  $I_A$ ,  $I_B$  and  $I_C$  respectively. Let  $d_1$ ,  $d_2$  and  $d_3$  be the spacings between the conductors as shown. Let us further assume that the loads are balanced i.e.  $I_A + I_B + I_C = 0$ . Consider the flux linkages with conductor  $A$ . There will be flux linkages with conductor  $A$  due to its own current and also due to the mutual inductance effects of  $I_B$  and  $I_C$

Flux linkages with conductor  $A$  due to its own current

$$= \frac{\mu_0 I_A}{2\pi} \left( \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right) \dots (i)$$

Flux linkages with conductor  $A$  due to current  $I_B$

$$= \frac{\mu_0 I_B}{2\pi} \int_{d_1}^\infty \frac{dx}{x} \dots (ii)$$

Flux linkages with conductor  $A$  due to current  $I_C$

$$= \frac{\mu_0 I_C}{2\pi} \int_{d_2}^\infty \frac{dx}{x} \dots (iii)$$

Total flux linkages with conductor  $A$  is

$$\psi_A = (i) + (ii) + (iii)$$

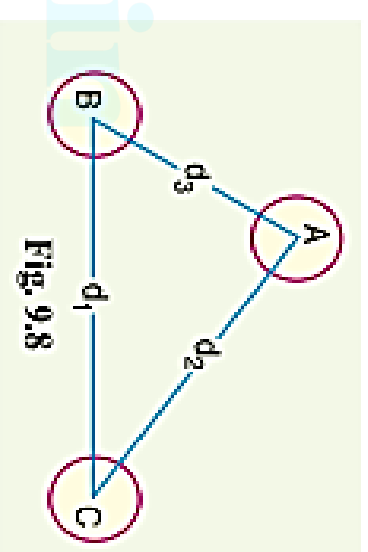


Fig. 9.8

Total flux linkages with conductor  $A$  is  $\psi_A$

$$\begin{aligned}\psi_A &= (i) + (ii) + (iii) \\ &= \frac{\mu_0 I_A}{2\pi} \left( \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right) + \frac{\mu_0 I_B}{2\pi} \int_{d_1}^\infty \frac{dx}{x} + \frac{\mu_0 I_C}{2\pi} \int_{d_2}^\infty \frac{dx}{x} \\ &= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} + \int_r^\infty \frac{dx}{x} \right) I_A + I_B \int_{d_1}^\infty \frac{dx}{x} + I_C \int_{d_2}^\infty \frac{dx}{x} \right] \\ &= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I_A - I_B \log_e d_3 - I_C \log_e d_2 + \log_e \infty (I_A + I_B + I_C) \right]\end{aligned}$$

As  $I_A + I_B + I_C = 0$ ,

$$\therefore \psi_A = \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I_A - I_B \log_e d_3 - I_C \log_e d_2 \right]$$

(ii) **Symmetrical spacing.** If the three conductors  $A$ ,  $B$  and  $C$  are placed symmetrically at the corners of an equilateral triangle of side  $d$ , then,  $d_1 = d_2 = d_3 = d$ . Under such conditions, the flux linkages with conductor  $A$  become :

$$\begin{aligned}\psi_A &= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I_A - I_B \log_e d - I_C \log_e d \right] \\ &= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I_A - (I_B + I_C) \log_e d \right] \\ &= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I_A + I_A \log_e d \right] \quad (\because I_B + I_C = -I_A) \\ &= \frac{\mu_0 I_A}{2\pi} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right] \text{ weber-turns/m}\end{aligned}$$

$$\text{Inductance of conductor } A, \quad L_A = \frac{\psi_A}{I_A} \text{ H/m} = \frac{\mu_0}{2\pi} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right] \text{ H/m}$$

$$= \frac{4\pi \times 10^{-7}}{2\pi} \left[ \frac{1}{4} + \log_e \frac{d}{r} \right] \text{ H/m}$$

$\therefore$

$$L_A = 10^{-7} \left[ 0.5 + 2 \log_e \frac{d}{r} \right] \text{ H/m}$$

Derived in a similar manner for conductors  $B$  and  $C$ .

for conductors  $B$  and  $C$ .

(ii) **Unsymmetrical spacing.** When 3-phase line conductors are not equidistant from each other, the conductor spacing is said to be unsymmetrical. Under such conditions, the flux linkages and inductance of each phase are not the same. A different inductance in each phase results in unequal voltage drops in the three phases even if the currents in the conductors are balanced. Therefore, the voltage at the receiving end will not be the same for all phases. In order that voltage drops are equal in all conductors, we generally interchange the positions of the conductors at regular intervals along the line so that each conductor occupies the original position of every other conductor over an equal distance. Such an exchange of positions is known as **transposition**. Fig. 9.9 shows the transposed line. The phase conductors are designated as *A*, *B* and *C* and the positions occupied are numbered 1, 2 and 3. The effect of transposition is that each conductor has the same average inductance.

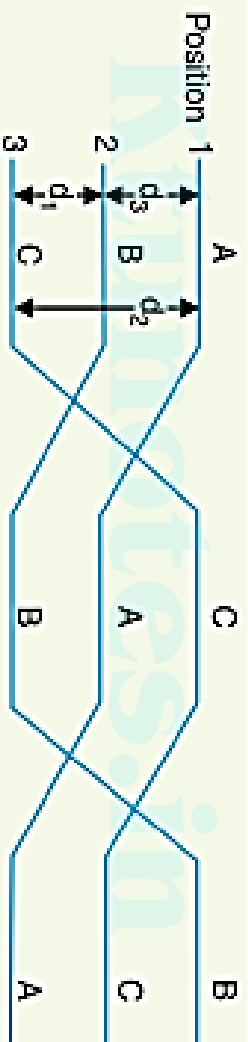


Fig. 9.9

Fig. 9.9 shows a 3-phase transposed line having unsymmetrical spacing. Let us assume that each of the three sections is 1 m in length. Let us further assume balanced conditions i.e.,  $I_A + I_B + I_C = 0$ . Let the line currents be :

$$I_A = I(1 + j0)$$

$$I_B = I(-0.5 - j0.866)$$

$$I_C = I(-0.5 + j0.866)$$

As proved above, the total flux linkages per metre length of conductor  $A$  is

$$\psi_A = \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I_A - I_B \log_e d_3 - I_C \log_e d_2 \right]$$

Putting the values of  $I_A$ ,  $I_B$  and  $I_C$ , we get,

$$\begin{aligned} \psi_A &= \frac{\mu_0}{2\pi} \left[ \left( \frac{1}{4} - \log_e r \right) I - I(-0.5 - j0.866) \log_e d_3 - I(-0.5 + j0.866) \log_e d_2 \right] \\ &= \frac{\mu_0}{2\pi} \left[ \frac{1}{4} I - I \log_e r + 0.5 I \log_e d_3 + j0.866 I \log_e d_3 + 0.5 I \log_e d_2 - j0.866 I \log_e d_2 \right] \\ &= \frac{\mu_0}{2\pi} \left[ \frac{1}{4} I - I \log_e r + 0.5 I (\log_e d_3 + \log_e d_2) + j0.866 I (\log_e d_3 - \log_e d_2) \right] \\ &= \frac{\mu_0}{2\pi} \left[ \frac{1}{4} I - I \log_e r + I^* \log_e \sqrt{d_2 d_3} + j0.866 I \log_e \frac{d_3}{d_2} \right] \\ &= \frac{\mu_0}{2\pi} \left[ \frac{1}{4} I + I \log_e \frac{\sqrt{d_2 d_3}}{r} + j0.866 I \log_e \frac{d_3}{d_2} \right] \\ &= \frac{\mu_0 I}{2\pi} \left[ \frac{1}{4} + \log_e \frac{\sqrt{d_2 d_3}}{r} + j0.866 \log_e \frac{d_3}{d_2} \right] \end{aligned}$$

$\therefore$  Inductance of conductor  $A$  is

$$\begin{aligned} L_A &= \frac{\psi_A}{I_A} = \frac{\psi_A}{I} \\ &= \frac{\mu_0}{2\pi} \left[ \frac{1}{4} + \log_e \frac{\sqrt{d_2 d_3}}{r} + j0.866 \log_e \frac{d_3}{d_2} \right] \end{aligned}$$

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$$* \quad 0.5 I (\log_e d_3 + \log_e d_2) = 0.5 I \log_e d_2 d_3 = I \log_e (d_2 d_3)^{0.5} = I \log_e \sqrt{d_2 d_3}$$

$$= \frac{4\pi \times 10^{-7}}{2\pi} \left[ \frac{1}{4} + \log_e \frac{\sqrt{d_2 d_3}}{r} + j 0.866 \log_e \frac{d_3}{d_2} \right] \text{ H/m}$$

$$= 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt{d_2 d_3}}{r} + j 1.732 \log_e \frac{d_3}{d_2} \right] \text{ H/m}$$

Similarly inductance of conductors B and C will be :

$$L_B = 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt{d_3 d_1}}{r} + j 1.732 \log_e \frac{d_1}{d_3} \right] \text{ H/m}$$

$$L_C = 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt{d_1 d_2}}{r} + j 1.732 \log_e \frac{d_2}{d_1} \right] \text{ H/m}$$

Inductance of each line conductor

$$= \frac{1}{3} (L_A + L_B + L_C)$$

$$= * \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r} \right] \times 10^{-7} \text{ H/m}$$

$$= \left[ 0.5 + 2 \log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r} \right] \times 10^{-7} \text{ H/m}$$

If we compare the formula of inductance of an unsymmetrically spaced transposed line with that of symmetrically spaced line, we find that inductance of each line conductor in the two cases will be equal if  $d = \sqrt[3]{d_1 d_2 d_3}$ . The distance  $d$  is known as *equivalent equilateral spacing* for unsymmetrically transposed line.

$$\begin{aligned}
 &= \frac{4\pi \times 10^{-7}}{2\pi} \left[ \frac{1}{4} + \log_e \frac{\sqrt{d_2 d_3}}{r} + j 0.866 \log_e \frac{d_3}{d_2} \right] \text{ H/m} \\
 &= 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt{d_2 d_3}}{r} + j 1.732 \log_e \frac{d_3}{d_2} \right] \text{ H/m}
 \end{aligned}$$

Similarly inductance of conductors B and C will be :

$$L_B = 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt{d_3 d_1}}{r} + j 1.732 \log_e \frac{d_1}{d_3} \right] \text{ H/m}$$

$$L_C = 10^{-7} \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt{d_1 d_2}}{r} + j 1.732 \log_e \frac{d_2}{d_1} \right] \text{ H/m}$$

Inductance of each line conductor

$$= \frac{1}{3} (L_A + L_B + L_C)$$

$$= \left[ \frac{1}{2} + 2 \log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r} \right] \times 10^{-7} \text{ H/m}$$

$$= \left[ 0.5 + 2 \log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r} \right] \times 10^{-7} \text{ H/m}$$

If we compare the formula of inductance of an unsymmetrically spaced transposed line with that of symmetrically spaced line, we find that inductance of each line conductor in the two cases will be equal if  $d = \sqrt[3]{d_1 d_2 d_3}$ . The distance  $d$  is known as *equivalent equilateral spacing* for unsymmetrically transposed line.

# Self GMD and Mutual GMD

The use of *self geometrical mean distance* (abbreviated as self-GMD) and *mutual geometrical mean distance* (mutual-GMD) simplifies the inductance calculations, particularly relating to multiconductor arrangements. The symbols used for these are respectively  $D_s$  and  $D_m$ . We shall briefly discuss these terms.

(i) **Self-GMD ( $D_s$ )**. In order to have concept of self-GMD (also sometimes called Geometrical mean radius ; GMR), consider the expression for inductance per conductor per metre already derived in Art. 9.5

$$\begin{aligned}\text{Inductance/conductor/m} &= 2 \times 10^{-7} \left( \frac{1}{4} + \log_e \frac{d}{r} \right) \\ &= 2 \times 10^{-7} \times \frac{1}{4} + 2 \times 10^{-7} \log_e \frac{d}{r} \quad \dots(i)\end{aligned}$$

$$\text{Inductance/conductor/m} = 2 \times 10^{-7} \log_e d/D_s^*$$

$$\text{where } D_s = \text{GMR or self-GMD} = 0.7788 r$$

# Self GMD and Mutual GMD

(ii) **Mutual-GMD.** The mutual-GMD is the geometrical mean of the distances from one conductor to the other and, therefore, must be between the largest and smallest such distance. In fact, mutual-GMD simply represents the equivalent geometrical spacing.

(a) The mutual-GMD between two conductors (assuming that spacing between conductors is large compared to the diameter of each conductor) is equal to the distance between their centres *i.e.*

$$D_m = \text{spacing between conductors} = d$$

(b) For a single circuit 3- $\phi$  line, the mutual-GMD is equal to the equivalent equilateral spacing *i.e.*,  $(d_1 d_2 d_3)^{1/3}$ .

$$D_m = (d_1 d_2 d_3)^{1/3}$$

(c) The principle of geometrical mean distances can be most profitably employed to 3- $\phi$  double circuit lines. Consider the conductor arrangement of the double circuit shown in Fig. 9-10. Suppose the radius of each conductor is  $r$ .

Self-GMD of conductor =  $0.7788 r$

Self-GMD of combination *aa'* is

$$D_{s1} = (*D_{aa} \times D_{aa'} \times D_{a'a} \times D_{a'a'})^{1/4}$$

Self-GMD of combination *bb'* is

$$D_{s2} = (D_{bb} \times D_{bb'} \times D_{b'b} \times D_{b'b'})^{1/4}$$

Self-GMD of combination *cc'* is

$$D_{s3} = (D_{cc} \times D_{cc'} \times D_{c'c} \times D_{c'c'})^{1/4}$$

Equivalent self-GMD of one phase

$$D_s = (D_{s1} \times D_{s2} \times D_{s3})^{1/3}$$

The value of  $D_s$  is the same for all the phases as each conductor has the same radius.

Mutual-GMD between phases *A* and *B* is

$$D_{AB} = (D_{ab} \times D_{ab'} \times D_{a'b} \times D_{a'b'})^{1/4}$$

Mutual-GMD between phases *B* and *C* is

$$D_{BC} = (D_{bc} \times D_{bc'} \times D_{b'c} \times D_{b'c'})^{1/4}$$

Mutual-GMD between phases *C* and *A* is

$$D_{CA} = (D_{ca} \times D_{ca'} \times D_{c'a} \times D_{c'a'})^{1/4}$$

$$\text{Equivalent mutual-GMD, } D_m = (D_{AB} \times D_{BC} \times D_{CA})^{1/3}$$

It is worthwhile to note that mutual GMD depends only upon the spacing and is substantially independent of the ex

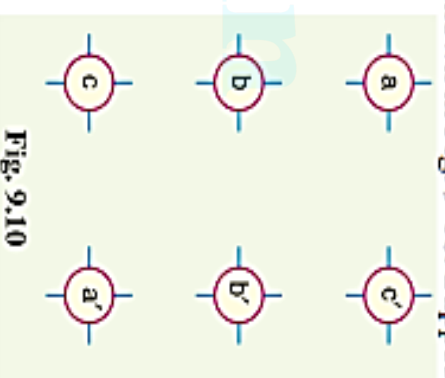


Fig. 9.10

# Inductance formula in terms of Self GMD and Mutual GMD

## (i) *Single phase line*

$$\text{Inductance/conductor/m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

where  $D_s = 0.7788 r$  and  $D_m = \text{Spacing between conductors} = d$

## (ii) *Single circuit 3- $\phi$ line*

$$\text{Inductance/phase/m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

where  $D_s = 0.7788 r$  and  $D_m = (d_1 d_2 d_3)^{1/3}$

## (iii) *Double circuit 3- $\phi$ line*

$$\text{Inductance/phase/m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

where  $D_s = (D_{s1} D_{s2} D_{s3})^{1/3}$  and  $D_m = (D_{AB} \times D_{BC} \times D_{CA})^{1/3}$

1. A single phase line has two parallel conductors 2 metres apart. The diameter of each conductor is 1.2 cm. Calculate the loop inductance per km of the line.

Spacing of conductors,  $d = 2 \text{ m} = 200 \text{ cm}$

Radius of conductor,  $r = 1.2/2 = 0.6 \text{ cm}$

Loop inductance per metre length of the line

$$\begin{aligned} &= 10^{-7} (1 + 4 \log_e d/r) \text{ H} \\ &= 10^{-7} (1 + 4 \log_e 200/0.6) \text{ H} \\ &= 24.23 \times 10^{-7} \text{ H} \end{aligned}$$

Loop inductance per km of the line

$$= 24.23 \times 10^{-7} \times 1000 = 24.23 \times 10^{-4} \text{ H} = \mathbf{2.423 \text{ mH}}$$

2. A single phase transmission line has two parallel conductors 3 m apart, the radius of each conductor being 1 cm. Calculate the loop inductance per km length of the line if the material of the conductor is (i) copper (ii) steel with relative permeability of 100.

Spacing of conductors,

$$d = 300 \text{ cm}$$

Radius of conductor,

$$r = 1 \text{ cm}$$

Loop inductance

$$= 10^{-7} (\mu_r + 4 \log_e d/r) \text{ H/m}$$

(i) With copper conductors,

$$\mu_r = 1$$

$$\therefore \text{ Loop inductance/m} = 10^{-7} (1 + 4 \log_e d/r) \text{ H} = 10^{-7} (1 + 4 \log_e 300/1) \text{ H}$$

$$= 23.8 \times 10^{-7} \text{ H}$$

Loop inductance/km

$$= 23.8 \times 10^{-7} \times 1000 = 2.38 \times 10^{-3} \text{ H} = \mathbf{2.38 \text{ mH}}$$

(ii) With steel conductors,  $\mu_r = 100$

$$\therefore \text{ Loop inductance/m} = 10^{-7} (100 + 4 \log_e 300/1) \text{ H} = 122.8 \times 10^{-7} \text{ H}$$

$$\text{Loop inductance/km} = 122.8 \times 10^{-7} \times 1000 = 12.28 \times 10^{-3} \text{ H} = \mathbf{12.28 \text{ mH}}$$

# Capacitance

$$C=Q/V$$

## Electric Potential

The electric potential at a point due to a charge is the work done in bringing a unit positive charge from infinity to that point. The concept of electric potential is extremely important for the determination of capacitance in a circuit since the latter is defined as the charge per unit potential. We shall now discuss in detail the electric potential due to some important conductor arrangements.

**(i) Potential at a charged single conductor.** Consider a long straight cylindrical conductor  $A$  of radius  $r$  metres. Let the conductor operate at such a potential ( $V_A$ ) that charge  $Q_A$  coulombs per metre exists on the conductor. It is desired to find the expression for  $V_A$ . The electric intensity  $E$  at a distance  $x$  from the centre of the conductor in air is given by:

$$E = \frac{Q_A}{2\pi x \epsilon_0} \text{ volts/m}$$

where

$Q_A$  = charge per metre length

$\epsilon_0$  = permittivity of free space

As  $x$  approaches infinity, the value of  $E$  approaches zero. Therefore, the potential difference between conductor  $A$  and infinity distant \* neutral plane is given by :

$$V_A = \int_0^\infty \frac{Q_A}{2\pi x \epsilon_0} dx = \frac{Q_A}{2\pi \epsilon_0} \int_0^\infty \frac{dx}{x}$$



Fig. 9.19

(ii) **Potential at a conductor in a group of charged conductors.** Consider a group of long straight conductors  $A, B, C$  etc. operating at potentials such that charges  $Q_A, Q_B, Q_C$  etc. coulomb per metre length exist on the respective conductors (see Fig. 9.20). Let us find the potential at  $A$  (i.e.  $V_A$ ) in this arrangement. Potential at  $A$  due to its own charge (i.e.  $Q_A$ )

$$= \int_r^\infty \frac{Q_A}{2\pi x \epsilon_0} dx \quad \dots(i)$$

Potential at conductor  $A$  due to charge  $Q_B$

$$= \int_{d_1}^\infty \frac{Q_B}{2\pi x \epsilon_0} dx \quad \dots(ii)$$

Potential at conductor  $A$  due to charge  $Q_C$

$$= \int_{d_2}^\infty \frac{Q_C}{2\pi x \epsilon_0} dx \quad \dots(iii)$$

Overall potential difference between conductor  $A$  and infinite neutral plane is

$$V_A = (i) + (ii) + (iii) + \dots$$

$$= \int_r^\infty \frac{Q_A}{2\pi x \epsilon_0} dx + \int_{d_1}^\infty \frac{Q_B}{2\pi x \epsilon_0} dx + \int_{d_2}^\infty \frac{Q_C}{2\pi x \epsilon_0} dx + \dots$$

$$= \frac{1}{2\pi \epsilon_0} \left[ Q_A (\log_e \infty - \log_e r) + Q_B (\log_e \infty - \log_e d_1) \right.$$

$$\left. + Q_C (\log_e \infty - \log_e d_2) + \dots \right]$$

$$= \frac{1}{2\pi \epsilon_0} \left[ Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_1} + Q_C \log_e \frac{1}{d_2} \right.$$

$$\left. + \log_e \infty (Q_A + Q_B + Q_C) + \dots \right]$$

Assuming balanced conditions i.e.,  $Q_A + Q_B + Q_C = 0$ , we have,

$$V_A = \frac{1}{2\pi \epsilon_0} \left[ Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_1} + Q_C \log_e \frac{1}{d_2} + \dots \right]$$

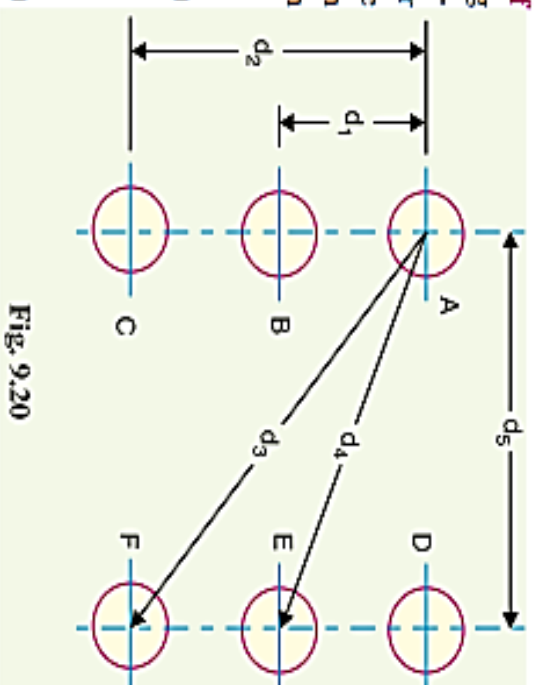


Fig. 9.20

# Capacitance of Single phase two wire line

- Lets consider a single phase two wire line.
- It consists of two parallel conductors A and B d meters apart in air.
- Let radius of each conductor be r and respective charge be +Q and -Q coulombs per meter.
- Then total pd between A and infinite neutral plane is

$$\begin{aligned}
 V_A &= \int_r^{\infty} \frac{Q}{2\pi x \epsilon_0} dx + \int_d^{\infty} \frac{-Q}{2\pi x \epsilon_0} dx \\
 &= \frac{Q}{2\pi \epsilon_0} \left[ \log_e \frac{\infty}{r} - \log_e \frac{\infty}{d} \right] \text{ volts} = \frac{Q}{2\pi \epsilon_0} \log_e \frac{d}{r} \text{ volts}
 \end{aligned}$$

Similarly, p.d. between conductor B and neutral "infinite" plane is

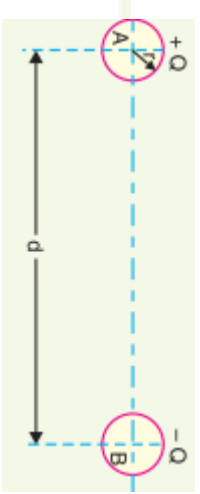
$$\begin{aligned}
 V_B &= \int_r^{\infty} \frac{-Q}{2\pi x \epsilon_0} dx + \int_d^{\infty} \frac{Q}{2\pi x \epsilon_0} dx \\
 &= \frac{-Q}{2\pi \epsilon_0} \left[ \log_e \frac{\infty}{r} - \log_e \frac{\infty}{d} \right] = \frac{-Q}{2\pi \epsilon_0} \log_e \frac{d}{r} \text{ volts}
 \end{aligned}$$

Both these potentials are w.r.t. the same neutral plane. Since the unlike charges attract each other, the potential difference between the conductors is

$$V_{AB} = 2V_A = \frac{2Q}{2\pi \epsilon_0} \log_e \frac{d}{r} \text{ volts}$$

$$\therefore \text{Capacitance, } C_{AB} = Q/V_{AB} = \frac{2Q}{2\pi \epsilon_0 \log_e \frac{d}{r}} \text{ F/m}$$

$$\therefore C_{AB} = \frac{\pi \epsilon_0}{\log_e \frac{d}{r}} \text{ F / m}$$



**Capacitance to neutral.** Equation (i) gives the capacitance between the conductors of a two-wire line [See Fig. 9.22]. Often it is desired to know the capacitance between one of the conductors and a neutral point between them. Since potential of the mid-point between the conductors is zero, the potential difference between each conductor and the ground or neutral is half the potential difference between the conductors. Thus the *capacitance to ground* or *capacitance to neutral* for the two-wire line is *twice* the line-to-line capacitance (capacitance between conductors as shown in Fig 9.23).

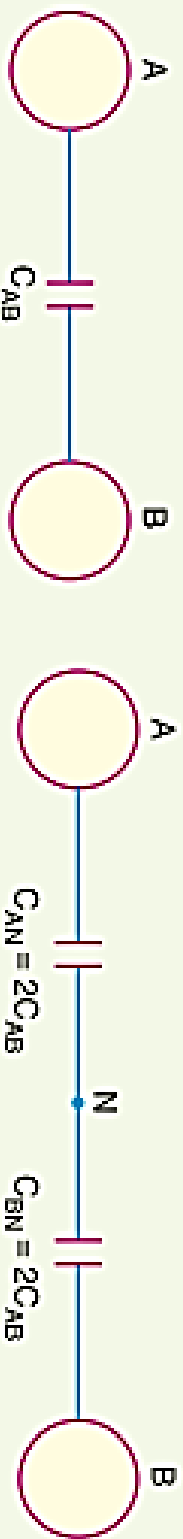


Fig. 9.22

Fig. 9.23

$$\therefore \text{Capacitance to neutral, } C_N = C_{AN} = C_{BN} = 2C_{AB}$$

$$\therefore C_N = \frac{2\pi\epsilon_0}{\log_e \frac{d}{r}} \text{ F / m} \quad \dots(ii)$$

The reader may compare eq. (ii) to the one for inductance. One difference between the equations for capacitance and inductance should be noted carefully. The radius in the equation for capacitance

$$\therefore \text{Capacitance to neutral, } C_N = C_{AN} = C_{BN} = 2C_{AB}$$

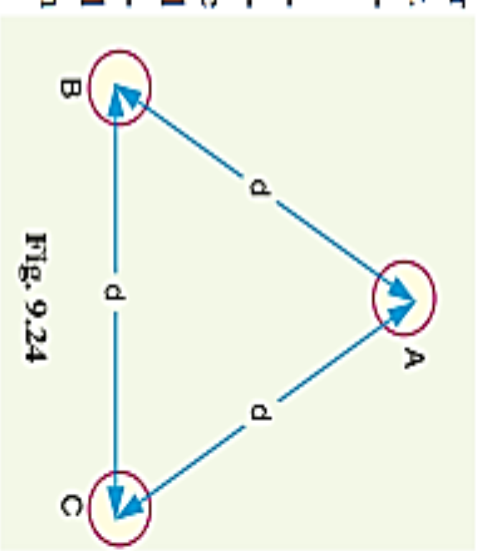
$$\therefore C_N = \frac{2\pi\epsilon_0}{\log_e \frac{d}{r}} \text{ F / m} \quad \dots(ii)$$

The reader may compare eq. (ii) to the one for inductance. One difference between the equations for capacitance and inductance should be noted carefully. The radius in the equation for capacitance

# Capacitance of 3 phase overhead line

In a 3-phase transmission line, the capacitance of each conductor is considered instead of capacitance from conductor to conductor. Here, again two cases arise viz., symmetrical spacing and unsymmetrical spacing.

(i) **Symmetrical Spacing.** Fig. 9.24 shows the three conductors  $A$ ,  $B$  and  $C$  of the 3-phase overhead transmission line having charges  $Q_A$ ,  $Q_B$  and  $Q_C$  per metre length respectively. Let the conductors be equidistant ( $d$  metres) from each other. We shall find the capacitance from line conductor to neutral in this symmetrically spaced line. Referring to Fig. 9.24, overall potential difference between conductor  $A$  and infinite neutral plane is given by (Refer to Art. 9.9);



$$\begin{aligned} V_A &= \int \frac{Q_A}{2\pi r \epsilon_0} dx + \int \frac{Q_B}{2\pi r \epsilon_0} dx + \int \frac{Q_C}{2\pi r \epsilon_0} dx \\ &= \frac{1}{2\pi \epsilon_0} \left[ Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d} + Q_C \log_e \frac{1}{d} \right] \\ &= \frac{1}{2\pi \epsilon_0} \left[ Q_A \log_e \frac{1}{r} + (Q_B + Q_C) \log_e \frac{1}{d} \right] \end{aligned}$$

Assuming balanced supply, we have,  $Q_A + Q_B + Q_C = 0$

$$\therefore Q_B + Q_C = -Q_A$$

$$\therefore V_A = \frac{1}{2\pi \epsilon_0} \left[ Q_A \log_e \frac{1}{r} - Q_A \log_e \frac{1}{d} \right] = \frac{Q_A}{2\pi \epsilon_0} \log_e \frac{d}{r} \text{ volts}$$

$\therefore$  Capacitance of conductor  $A$  w.r.t neutral,

$$C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi \epsilon_0} \log_e \frac{d}{r}} \text{ F / m} = \frac{2\pi \epsilon_0}{\log_e \frac{d}{r}} \text{ F / m}$$

$$\therefore C_r = \frac{2\pi \epsilon_0}{\log_e \frac{d}{r}} \text{ F / m}$$

(ii) **Unsymmetrical spacing.** Fig. 9.25 shows a 3-phase transposed line having unsymmetrical spacing. Let us assume balanced conditions i.e.  $Q_A + Q_B + Q_C = 0$ .

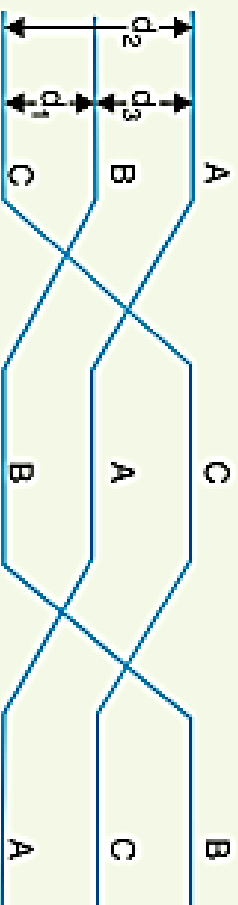


Fig. 9.25

Considering all the three sections of the transposed line for phase A,

$$\text{Potential of 1st position, } V_1 = \frac{1}{2\pi\epsilon_0} \left( Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_3} + Q_C \log_e \frac{1}{d_2} \right)$$

$$\text{Potential of 2nd position, } V_2 = \frac{1}{2\pi\epsilon_0} \left( Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_1} + Q_C \log_e \frac{1}{d_3} \right)$$

$$\text{Potential of 3rd position, } V_3 = \frac{1}{2\pi\epsilon_0} \left( Q_A \log_e \frac{1}{r} + Q_B \log_e \frac{1}{d_2} + Q_C \log_e \frac{1}{d_1} \right)$$

Average voltage on conductor A is

$$\begin{aligned} V_A &= \frac{1}{3} (V_1 + V_2 + V_3) \\ &= \frac{1}{3 \times 2\pi\epsilon_0} \left[ Q_A \log_e \frac{1}{r} + (Q_B + Q_C) \log_e \frac{1}{d_1 d_2 d_3} \right] \end{aligned}$$

As  $Q_A + Q_B + Q_C = 0$ , therefore,  $Q_B + Q_C = -Q_A$

$$\therefore V_A = \frac{1}{6\pi\epsilon_0} \left[ Q_A \log_e \frac{1}{r} - Q_A \log_e \frac{1}{d_1 d_2 d_3} \right]$$

$$\begin{aligned}
 &= \frac{Q_A}{6\pi\epsilon_0} \log_e \frac{d_1 d_2 d_3}{r^3} \\
 &= \frac{1}{3} \times \frac{Q_A}{2\pi\epsilon_0} \log_e \frac{d_1 d_2 d_3}{r^3} \\
 &= \frac{Q_A}{2\pi\epsilon_0} \log_e \left( \frac{d_1 d_2 d_3}{r^3} \right)^{1/3} \\
 &= \frac{Q_A}{2\pi\epsilon_0} \log_e \frac{(d_1 d_2 d_3)^{1/3}}{r}
 \end{aligned}$$

∴ Capacitance from conductor to neutral is

$$C_A = \frac{Q_A}{V_A} = \frac{2\pi\epsilon_0}{\log_e \frac{\sqrt[3]{d_1 d_2 d_3}}{r}} \text{ F/m}$$



# Tutorial 2

1. A single-phase transmission line has two parallel conductors 3 metres apart, radius of each conductor being 1 cm. Calculate the capacitance of the line per km. Given that  $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ .

Conductor radius,

$$r = 1 \text{ cm}$$

Spacing of conductors,

$$d = 3 \text{ m} = 300 \text{ cm}$$

Capacitance of the line

$$\begin{aligned} &= \frac{\pi \epsilon_0}{\log_e d/r} \text{ F/m} = \frac{\pi \times 8.854 \times 10^{-12}}{\log_e 300/1} \text{ F/m} \\ &= 0.4875 \times 10^{-11} \text{ F/m} = 0.4875 \times 10^{-8} \text{ F/km} \\ &= \mathbf{0.4875 \times 10^{-2} \mu\text{F/km}} \end{aligned}$$



2.

A 3-phase overhead transmission line has its conductors arranged at the corners of an equilateral triangle of 2 m side. Calculate the capacitance of each line conductor per km. Given that diameter of each conductor is 1.25 cm.

Conductor radius,  $r = 1.25/2 = 0.625 \text{ cm}$

Spacing of conductors,  $d = 2 \text{ m} = 200 \text{ cm}$

Capacitance of each line conductor

$$\begin{aligned}
 &= \frac{2 \pi \epsilon_0}{\log_e d/r} \text{ F/m} = \frac{2 \pi \times 8.854 \times 10^{-12}}{\log_e 200/0.625} \text{ F/m} \\
 &= 0.0096 \times 10^{-9} \text{ F/m} = 0.0096 \times 10^{-6} \text{ F/km} = \mathbf{0.0096 \mu\text{F/km}}
 \end{aligned}$$

# Classification of Overhead Transmission Lines

Depending upon the manner in which capacitance is taken into account, the overhead transmission lines are classified as :

## **1. *Short transmission lines.***

- An overhead transmission line upto about 50 km and the line voltage is comparatively low ( $< 20 \text{ kV}$ )
- Due to smaller length and lower voltage, the capacitance effects are small and hence can be neglected.
- Therefore, while studying the performance of a short transmission line, only resistance and inductance of the line are taken into account.

# Classification of Overhead Transmission Lines

## 2. *Medium transmission lines.*

- An overhead transmission line is about 50- 150 km and the line voltage is moderately high ( $>20 \text{ kV} < 100 \text{ kV}$ ),
- Due to sufficient length and voltage of the line, the capacitance effects are taken into account.
- For purposes of calculations, the distributed capacitance of the line is divided and lumped in the form of condensers shunted across the line at one or more points.

## 3. *Long transmission lines.*

- An overhead transmission line is more than 150 km and line voltage is very high ( $> 100 \text{ kV}$ ), it is considered as a long transmission line.
- For the treatment of such a line, the line constants are considered uniformly distributed over the whole length of the line and rigorous methods are employed for solution.

# Important Terms

(ii) **Transmission efficiency.** The power obtained at the receiving end of a transmission line is generally less than the sending end power due to losses in the line resistance.

*The ratio of receiving end power to the sending end power of a transmission line is known as the transmission efficiency of the line i.e.*

$$\begin{aligned}\% \text{ age Transmission efficiency, } \eta_T &= \frac{\text{Receiving end power}}{\text{Sending end power}} \times 100 \\ &= \frac{V_R I_R \cos \phi_R}{V_S I_S \cos \phi_S} \times 100\end{aligned}$$

where  $V_R, I_R$  and  $\cos \phi_R$  are the receiving end voltage, current and power factor while  $V_S, I_S$  and  $\cos \phi_S$  are the corresponding values at the sending end.

# Performance of Single phase Short Transmission lines

As stated earlier, the effects of line capacitance are neglected for a short transmission line. Therefore, while studying the performance of such a line, only resistance and inductance of the line are taken into account. The equivalent circuit of a single phase short transmission line is shown in Fig. 10.1 (i). Here, the total line resistance and inductance are shown as concentrated or lumped instead of being distributed. The circuit is a simple a.c. series circuit.

Let

$I$  = load current

$R$  = loop resistance i.e., resistance of both conductors

$X_L$  = loop reactance

$V_R$  = receiving end voltage

$\cos \phi_R$  = receiving end power factor (lagging)

$V_S$  = sending end voltage

$\cos \phi_S$  = sending end power factor

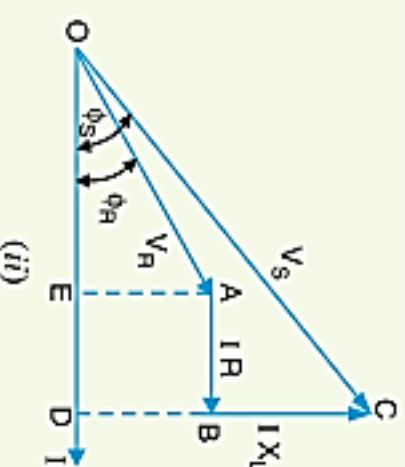
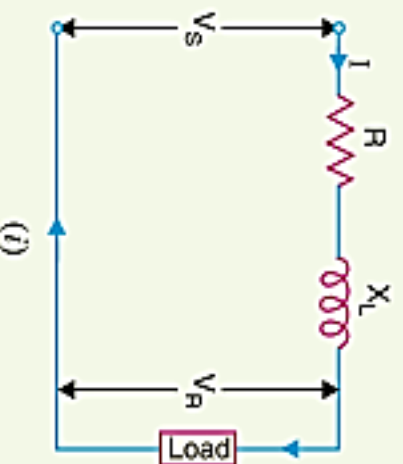


Fig. 10.1

The \*phasor diagram of the line for lagging load power factor is shown in Fig. 10.1 (ii). From the right angled triangle  $ODC$ , we get,

$$(OC)^2 = (OD)^2 + (DC)^2$$

$$V_s^2 = (OE + ED)^2 + (DB + BC)^2$$

$$= (V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX_L)^2$$

or

$$\therefore V_s = \sqrt{(V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX_L)^2}$$

$$(i) \text{ \%age Voltage regulation} = \frac{V_s - V_R}{V_R} \times 100$$

$$(ii) \text{ Sending end } p.f., \cos \phi_s = \frac{OD}{OC} = \frac{V_R \cos \phi_R + IR}{V_s}$$

$$(iii) \text{ Power delivered} = V_R I_R \cos \phi_R$$

$$\text{Line losses} = I^2 R$$

$$\text{Power sent out} = V_R I_R \cos \phi_R + I^2 R$$

$$\% \text{age Transmission efficiency} = \frac{\text{Power delivered}}{\text{Power sent out}} \times 100$$

$$= \frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + I^2 R} \times 100$$

An approximate expression for the sending end voltage  $V_s$  can be obtained as follows. Draw perpendicular from B and C on OA produced as shown in Fig. 10.2. Then OC is nearly equal to OF, i.e.,

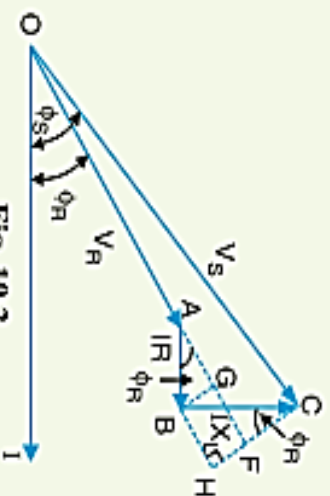


Fig. 10.2

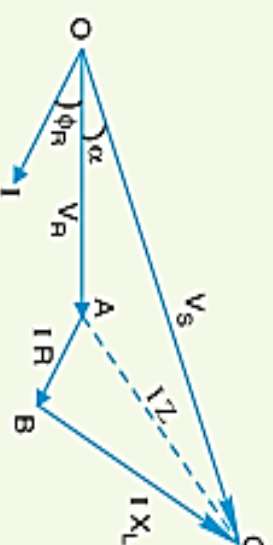


Fig. 10.3

$$OC = OF = OA + AF = OA + AG + GF$$

$$= OA + AG + BH$$

$\therefore$

$$V_s = V_R + IR \cos \phi_R + IX_L \sin \phi_R$$

**Solution in complex notation.** It is often convenient and profitable to make the line calculations in complex notation.

Taking  $\vec{V}_R$  as the reference phasor, draw the phasor diagram as shown in Fig 10.3. It is clear that  $\vec{V}_S$  is the phasor sum of  $\vec{V}_R$  and  $\vec{I}\vec{Z}$ .

$$\begin{aligned}\overset{*}{V}_R &= V_R + j0 \\ \vec{I} &= \vec{I} \angle -\phi_R = I (\cos \phi_R - j \sin \phi_R) \\ \vec{Z} &= R + jX_L \\ \vec{V}_S &= \vec{V}_R + \vec{I}\vec{Z} \\ &= (V_R + j0) + I (\cos \phi_R - j \sin \phi_R) (R + jX_L)\end{aligned}$$

$$= (V_R + IR \cos \phi_R + IX_L \sin \phi_R) + j (IX_L \cos \phi_R - IR \sin \phi_R)$$

$$\therefore V_S = \sqrt{(V_R + IR \cos \phi_R + IX_L \sin \phi_R)^2 + (IX_L \cos \phi_R - IR \sin \phi_R)^2}$$

The second term under the root is quite small and can be neglected with reasonable accuracy. Therefore, approximate expression for  $V_S$  becomes :

$$V_S = V_R + IR \cos \phi_R + IX_L \sin \phi_R$$

The following points may be noted :

- (i) The approximate formula for  $V_S$  ( $= V_R + IR \cos \phi_R + IX_L \sin \phi_R$ ) gives fairly correct results for lagging power factors. However, appreciable error is caused for leading power factors. Therefore, approximate expression for  $V_S$  should be used for lagging p.f. only.
- (ii) The solution in complex notation is in more presentable form.

# Three phase short transmission lines

For reasons associated with economy, transmission of electric power is done by 3-phase system. This system may be regarded as consisting of three single phase units, each wire transmitting one-third of the total power. As a matter of convenience, we generally analyse 3-phase system by considering

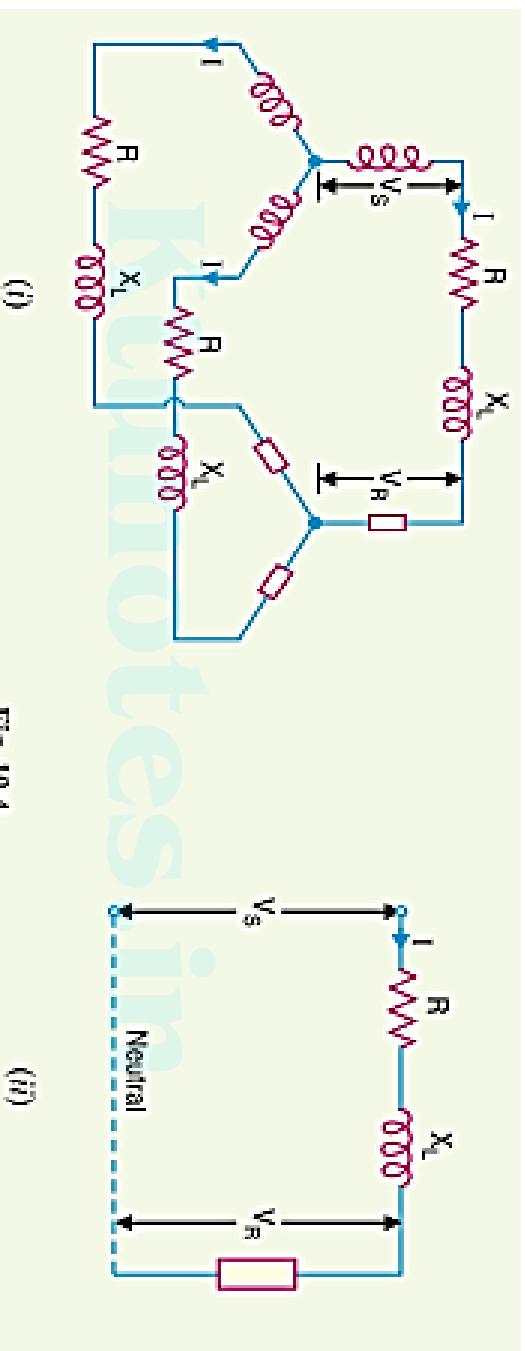


Fig. 10.4

\*one phase only. Therefore, expression for regulation, efficiency etc. derived for a single phase line can also be applied to a 3-phase system. Since only one phase is considered, phase values of 3-phase system should be taken. Thus,  $V_s$  and  $V_R$  are the phase voltages, whereas  $R$  and  $X_L$  are the resistance and inductive reactance per phase respectively.

Fig. 10.4 (i) shows a Y-connected generator supplying a balanced Y-connected load through a transmission line. Each conductor has a resistance of  $R \Omega$  and inductive reactance of  $X_L \Omega$ . Fig. 10.4 (ii) shows one phase separately. The calculations can now be made in the same way as for a single phase line.

# Effect of load pf on Voltage Regulation and efficiency

The regulation and efficiency of a transmission line depend to a considerable extent upon the power factor of the load.

**1. Effect on regulation.** The expression for voltage regulation of a short transmission line is given by :

$$\% \text{age Voltage regulation} = \frac{I R \cos \phi_R + I X_L \sin \phi_R}{V_R} \times 100 \quad (\text{for lagging p.f.})$$

$$\% \text{age Voltage regulation} = \frac{I R \cos \phi_R - I X_L \sin \phi_R}{V_R} \times 100 \quad (\text{for leading p.f.})$$

The following conclusions can be drawn from the above expressions :

- (i) When the load p.f. is lagging or unity or such leading that  $I R \cos \phi_R > I X_L \sin \phi_R$ , then voltage regulation is positive i.e., receiving end voltage  $V_R$  will be less than the sending end voltage  $V_S$ .
- (ii) For a given  $V_R$  and  $I$ , the voltage regulation of the line increases with the decrease in p.f. for lagging loads.
- (iii) When the load p.f. is leading to this extent that  $I X_L \sin \phi_R > I R \cos \phi_R$ , then voltage regulation is negative i.e. the receiving end voltage  $V_R$  is more than the sending end voltage  $V_S$ .
- (iv) For a given  $V_R$  and  $I$ , the voltage regulation of the line decreases with the decrease in p.f. for leading loads.

**2. Effect on transmission efficiency.** The power delivered to the load depends upon the power factor.

$$P = V_R * I \cos \phi_R \quad (\text{For 1-phase line})$$

$$\therefore I = \frac{P}{V_R \cos \phi_R}$$

$$P = 3 V_R I \cos \phi_R \quad (\text{For 3-phase line})$$

$$\therefore I = \frac{P}{3V_R \cos \phi_R}$$

It is clear that in each case, for a given amount of power to be transmitted ( $P$ ) and receiving end voltage



Power Factor Regulator



Power Factor Meter

( $V_R$ ), the load current  $I$  is inversely proportional to the load p.f.  $\cos \phi_R$ . Consequently, with the decrease in load p.f., the load current and hence the line losses are increased. This leads to the conclusion that transmission efficiency of a line decreases with the decrease in load p.f. and *vice-versa*,

# Tutorial 3

1. A single phase overhead transmission line delivers 1100 kW at 33 kV at 0.8 p.f. lagging. The total resistance and inductive reactance of the line are 10Ω and 15Ω respectively. Determine : (i) sending end voltage (ii) sending end power factor and (iii) transmission efficiency.

Load power factor,  $\cos \phi_R = 0.8$  lagging

Total line impedance,  $\vec{Z} = R + jX_L = 10 + j15$

Receiving end voltage,  $V_R = 33 \text{ kV} = 33,000 \text{ V}$

$$\therefore \text{Line current, } I = \frac{kW \times 10^3}{V_R \cos \phi_R} = \frac{1100 \times 10^3}{33,000 \times 0.8} = 41.67 \text{ A}$$

$$\text{As } \cos \phi_R = 0.8 \quad \therefore \sin \phi_R = 0.6$$

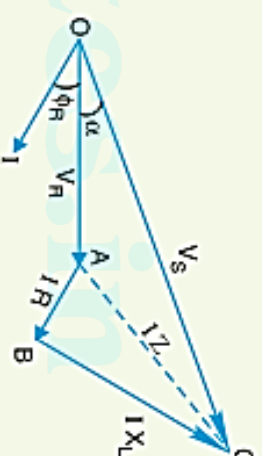
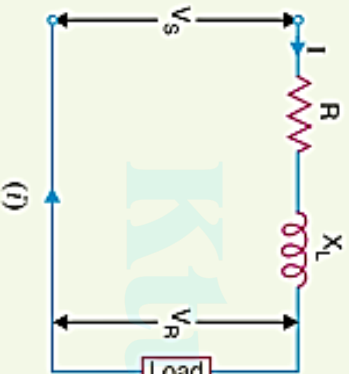


Fig. 10.5

The equivalent circuit and phasor diagram of the line are shown in Figs. 10.5 (i) and 10.5 (ii) respectively. Taking receiving end voltage  $\vec{V}_R$  as the reference phasor,

$$\vec{V}_R = V_R + j0 = 33000 \text{ V}$$

$$\vec{I} = I (\cos \phi_R - j \sin \phi_R)$$

$$= 41.67 (0.8 - j0.6) = 33.33 - j25$$

$$(i) \text{ Sending end voltage, } \vec{V}_S = \vec{V}_R + \vec{I} \vec{Z}$$

$$= 33,000 + (33.33 - j25.0) (10 + j15)$$

$$= 33,000 + 333.3 - j250 + j500 + 375$$

$$= 33,708.3 + j250$$

$\therefore$

Mag



(ii) Angle between  $\vec{V}_S$  and  $\vec{V}_R$  is

$$\alpha = \tan^{-1} \frac{2.50}{33,708.3} = \tan^{-1} 0.0074 = 0.42^\circ$$

$\therefore$  Sending end power factor angle is

$$\phi_S = \phi_R + \alpha = 36.87^\circ + 0.42^\circ = 37.29^\circ$$

$\therefore$  Sending end p.f.,  $\cos \phi_S = \cos 37.29^\circ = \mathbf{0.7956 \text{ lagging}}$

(iii) Line losses  $= I^2 R = (41.67)^2 \times 10 = 17,364 \text{ W} = 17.364 \text{ kW}$

Output delivered  $= 1100 \text{ kW}$

$$\text{Power sent} = 1100 + 17.364 = 1117.364 \text{ kW}$$

$\therefore$  Transmission efficiency  $= \frac{\text{Power delivered}}{\text{Power sent}} \times 100 = \frac{1100}{1117.364} \times 100 = \mathbf{98.44\%}$

**Note.**  $V_S$  and  $\phi_S$  can also be calculated as follows :

$$V_S = V_R + IR \cos \phi_R + IX_L \sin \phi_R \text{ (approximately)}$$

$$= 33,000 + 41.67 \times 10 \times 0.8 + 41.67 \times 15 \times 0.6$$

$$= 33,000 + 333.36 + 375.03$$

$= 33708.39 \text{ V}$  which is approximately the same as above

$$\begin{aligned} \cos \phi_S &= \frac{V_R \cos \phi_R + IR}{V_S} = \frac{33,000 \times 0.8 + 41.67 \times 10}{33,708.39} = \frac{26,816.7}{33,708.39} \\ &= 0.7958 \end{aligned}$$

# Medium Transmission Lines

- The medium transmission lines have sufficient length (50-150 km) and usually operate at voltages greater than 20 kV, so effects of capacitance cannot be neglected.
- However, in order to make the calculations simple, the line capacitance is assumed to be lumped or concentrated in the form of capacitors shunted across the line at one or more points.
- The most commonly used methods
  1. End condenser method
  2. Nominal  $T$  method
  3. Nominal  $\pi$  method.

# End Condenser Method

This method of localising the line capacitance at the load end overestimates the effects of capacitance

Let  $I_R$  = load current per phase

$R$  = resistance per phase

$X_L$  = inductive reactance per phase

$C$  = capacitance per phase

$\cos \phi_R$  = receiving end power factor (*lagging*)

$V_S$  = sending end voltage per phase

The \*phasor diagram for the circuit is shown in Fig 10.9.

Taking the receiving end voltage  $\vec{V}_R$  as the reference phasor,

we have,  $\vec{V}_R = V_R + j0$

Load current,  $\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R)$

Capacitive current,  $\vec{I}_C = j \vec{V}_R \omega C = j 2 \pi f C \vec{V}_R$

The sending end current  $\vec{I}_S$  is the phasor sum of load current

$\vec{I}_R$  and capacitive current  $\vec{I}_C$  i.e.,

$$\begin{aligned} \vec{I}_S &= \vec{I}_R + \vec{I}_C \\ &= I_R (\cos \phi_R - j \sin \phi_R) + j 2 \pi f C V_R \\ &= I_R \cos \phi_R + j (-I_R \sin \phi_R + 2 \pi f C V_R) \end{aligned}$$

Voltage drop/phase  $= \vec{I}_S \vec{Z} = \vec{I}_S (R + jX_L)$

Sending end voltage,  $\vec{V}_S = \vec{V}_R + \vec{I}_S \vec{Z} = \vec{V}_R + \vec{I}_S (R + jX_L)$

Thus, the magni

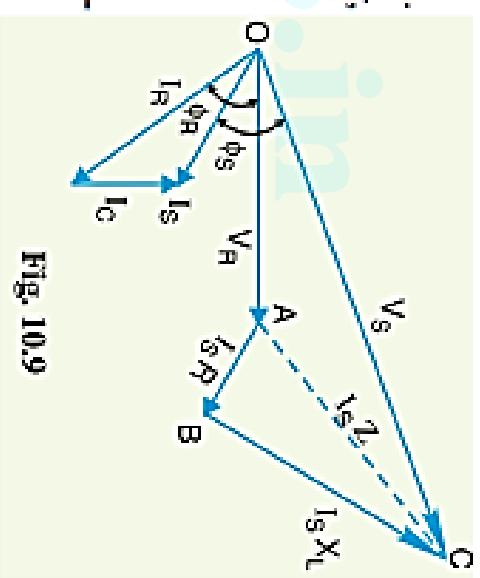
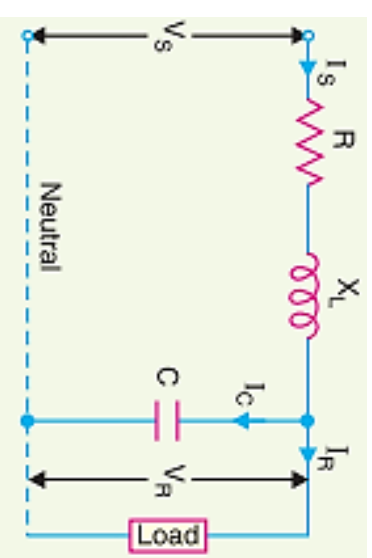


Fig. 10.9

$$\% \text{ Voltage regulation} = \frac{V_S - V_R}{V_R} \times 100$$

$$\% \text{ Voltage transmission efficiency} = \frac{\text{Power delivered / phase}}{\text{Power delivered / phase} + \text{losses / phase}} \times 100$$

$$= \frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + I_S^2 R} \times 100$$

*Limitations.* Although end condenser method for the solution of medium lines is simple to work out calculations, yet it has the following drawbacks :

- (i) There is a considerable error (about 10%) in calculations because the distributed capacitance has been assumed to be lumped or concentrated.
- (ii) This method overestimates the effects of line capacitance.

# Tutorial 4

1. A (medium) single phase transmission line 100 km long has the following constants : Resistance/km =  $0.25 \Omega$  ; Reactance/km =  $0.8 \Omega$  Susceptance/km =  $14 \times 10^{-6}$  siemen ; Receiving end line voltage = 66,000 V Assuming that the total capacitance of the line is localised at the receiving end alone, determine

- (i) the sending end current
- (ii) the sending end voltage
- (iii) regulation and
- (iv) supply power factor.

The line is delivering 15,000 kW at 0.8 power factor lagging. Draw the phasor diagram to illustrate your calculations.

Total resistance,  $R = 0.25 \times 100 = 25 \Omega$

Total reactance,  $X_L = 0.8 \times 100 = 80 \Omega$

Total susceptance,  $Y = 14 \times 10^{-6} \times 100 = 14 \times 10^{-4} S$

Receiving end voltage,  $V_R = 66,000 V$

$\therefore$  Load current,  $I_R = \frac{15,000 \times 10^3}{66,000 \times 0.8} = 284 A$

$\cos \phi_R = 0.8$ ;  $\sin \phi_R = 0.6$

Taking receiving end voltage as the reference phasor [see Fig. 10.10 (ii)], we have,

$$\vec{V}_R = V_R + j0 = 66,000 V$$

Load current,  $\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R) = 284 (0.8 - j 0.6) = 227 - j 170$

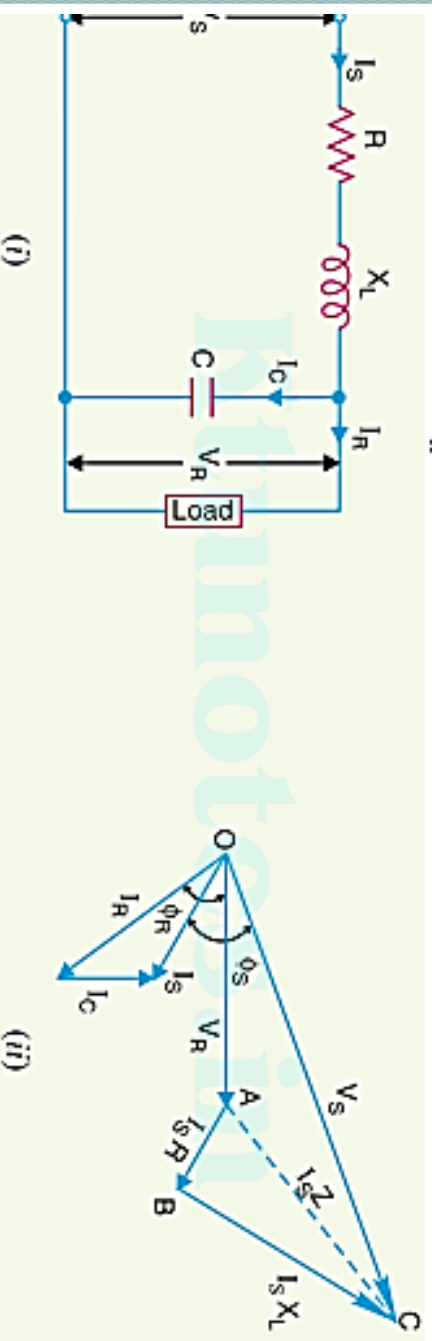


Fig. 10.10

Capacitive current,  $\vec{I}_C = jY \times V_R = j 14 \times 10^{-4} \times 66000 = j 92$

(i) Sending end current,  $\vec{I}_S = \vec{I}_R + \vec{I}_C = (227 - j 170) + j 92$   
 $= 227 - j 78$

... (i)

Magnitude of  $I_S = \sqrt{(227)^2 + (78)^2} = 240 A$

(ii) Voltage drop  
 $= \vec{I}_S \vec{Z} = \vec{I}_S (R + j X_L) = (227 - j 78) (25 + j 80)$   
 $= 5,675 + j 18,160 - j 1950 + 6240$



Sending end voltage,  $\vec{V}_S = \vec{V}_R + \vec{I}_S \vec{Z} = 66,000 + 11,915 + j 16,210$   
 $= 77,915 + j 16,210$

Magnitude of  $V_S = \sqrt{(77915)^2 + (16210)^2} = \mathbf{79583V}$

(iii) % Voltage regulation  $= \frac{V_S - V_R}{V_R} \times 100 = \frac{79,583 - 66,000}{66,000} \times 100 = \mathbf{20.58\%}$

(iv) Referring to exp. (i), phase angle between  $\vec{V}_R$  and  $\vec{I}_R$  is :

$$\theta_1 = \tan^{-1} \frac{78}{227} = \tan^{-1} (-0.3436) = -18.96^\circ$$

Referring to exp. (ii), phase angle between  $\vec{V}_R$  and  $\vec{V}_S$  is :

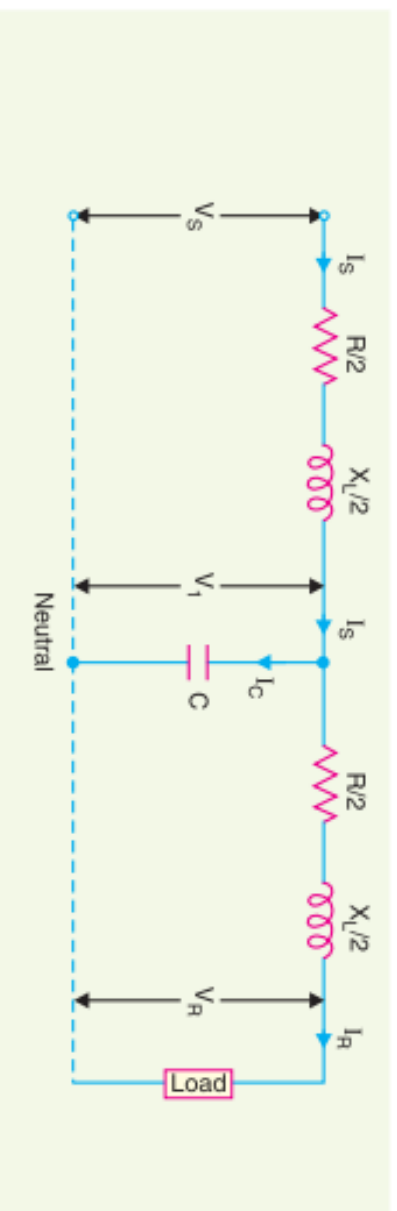
$$\theta_2 = \tan^{-1} \frac{16210}{77915} = \tan^{-1} (0.2036) = 11.50^\circ$$

$$\therefore \text{Supply power factor angle, } \phi_S = 18.96^\circ + 11.50^\circ = 30.46^\circ$$

$$\therefore \text{Supply p.f.} = \cos \phi_S = \cos 30.46^\circ = \mathbf{0.86 \text{ lag}}$$

# Nominal T Method

- In this method, the whole line capacitance is assumed to be concentrated at the middle point of the line and half the line resistance and reactance are lumped on its either side as shown in Fig.
- Therefore, in this arrangement, full charging current flows over half the line.
- In Fig one phase of 3- phase transmission line is shown as it is advantageous to work in phase instead of line-to-line values.



Let  $I_R$  = load current per phase ;

$R$  = resistance per phase

$X_L$  = inductive reactance per phase ;

$C$  = capacitance per phase

$\cos \phi_R$  = receiving end power factor (*lagging*) ;

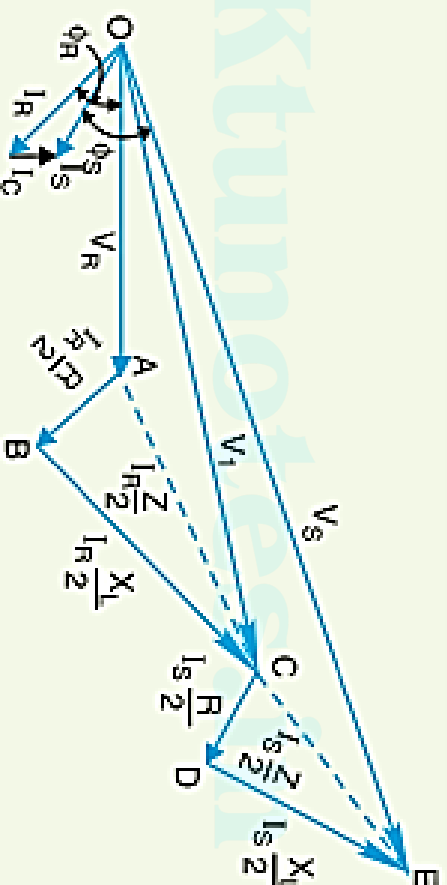
$V_S$  = sending end voltage/phase

$V_I$  = voltage across capacitor  $C$

The \*phasor diagram for the circuit is shown in Fig. 10.12. Taking the receiving end voltage  $\vec{V}_R$  as the reference phasor, we have,

Receiving end voltage,  $\vec{V}_R = V_R + j0$

Load current,  $\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R)$



Voltage across  $C$ ,  $\vec{V}_I = \vec{V}_R + \vec{I}_R \vec{Z} / 2$

$$= V_R + I_R (\cos \phi_R - j \sin \phi_R) \left( \frac{R}{2} + j \frac{X_L}{2} \right)$$

Capacitive current,  $\vec{I}_C = j \omega C \vec{V}_I = j 2\pi f C \vec{V}_I$

Sending end current,  $\vec{I}_S = \vec{I}_R + \vec{I}_C$

So  $\vec{V}_S = \vec{V}_R + \vec{I}_S \vec{Z} = \vec{V}_R + \vec{I}_R \vec{Z} + j \left( \frac{R}{2} + j \frac{X_L}{2} \right) \vec{V}_I$

# Tutorial 5

*1. A 3-phase, 50-Hz overhead transmission line 100 km long has the following constants :*

$$\text{Resistance/km/phase} = 0.1 \, \Omega$$

$$\text{Inductive reactance/km/phase} = 0.2 \, \Omega$$

$$\text{Capacitive susceptance/km/phase} = 0.04 \times 10^{-4} \text{ siemen}$$

*Determine (i) the sending end current (ii) sending end voltage (iii) sending end power factor and (iv) transmission efficiency when supplying a balanced load of 10,000 kW at 66 kV, p.f. 0.8 lagging. Use nominal T method.*



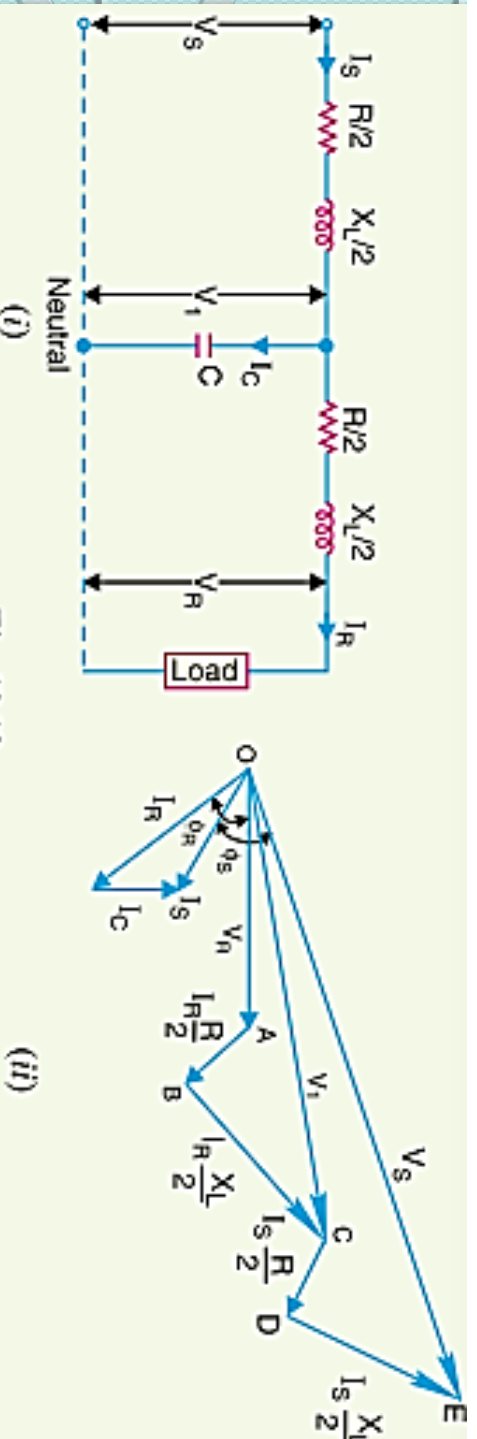


Fig. 10.13

Total resistance/phase,

$$R = 0.1 \times 100 = 10 \, \Omega$$

Total reactance/phase,

$$X_L = 0.2 \times 100 = 20 \, \Omega$$

Capacitive susceptance,

$$Y = 0.04 \times 10^{-4} \times 100 = 4 \times 10^{-4} \, \text{S}$$

Receiving end voltage/phase,

$$V_R = 66,000/\sqrt{3} = 38,105 \, \text{V}$$

Load current,

$$I_R = \frac{10,000 \times 10^3}{\sqrt{3} \times 66 \times 10^3 \times 0.8} = 109 \, \text{A}$$

$$\cos \phi_R = 0.8; \sin \phi_R = 0.6$$

Impedance per phase,  $\vec{Z} = R + jX_L = 10 + j20$

(i) Taking receiving end voltage as the reference phasor [see Fig. 10.13 (ii)], we have,

Receiving end voltage,  $\vec{V}_R = V_R + j0 = 38,105 \, \text{V}$

Load current,  $\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R) = 109 (0.8 - j0.6) = 87.2 - j65.4$

Voltage across  $C$ ,  $\vec{V}_1 = \vec{V}_R + \vec{I}_R \vec{Z}/2 = 38,105 + (87.2 - j65.4) (5 + j10)$   
 $= 38,105 + 436 + j872 - j327 + 654 = 39,195 + j545$

Charging current,  $\vec{I}_C = jY\vec{V}_1 = j4 \times 10^{-4}(39,195 + j545) = -0.218 + j15.6$

Sending end current,  $\vec{I}_S = \vec{I}_R + \vec{I}_C = (87.2 - j65.4) + (-0.218 + j15.6)$

$$= 87.0 - j49.8 = 100 \angle -29^\circ 47' \text{ A}$$

$\therefore$  Sending end current = **100 A**

(ii) Sending end voltage,  $\vec{V}_S = \vec{V}_1 + \vec{I}_S \vec{Z}/2 = (39,195 + j545) + (87.0 - j49.8)(5 + j10)$

$$= 39,195 + j545 + 434.9 + j870 - j249 + 498$$

$$= 40128 + j1170 = 40145 \angle 1^\circ 40' \text{ V}$$

$\therefore$  Line value of sending end voltage

$$= 40145 \times \sqrt{3} = 69\,533 \text{ V} = \mathbf{69.533 \text{ kV}}$$

(iii) Referring to phasor diagram in Fig. 10.14,

$$\theta_1 = \text{angle between } \vec{V}_R \text{ and } \vec{V}_S = 1^\circ 40'$$

$$\theta_2 = \text{angle between } \vec{V}_R \text{ and } \vec{I}_S = 29^\circ 47'$$

$\therefore$   $\phi_S$  = angle between  $\vec{V}_S$  and  $\vec{I}_S$

$$= \theta_1 + \theta_2 = 1^\circ 40' + 29^\circ 47' = 31^\circ 27'$$

$\therefore$  Sending end power factor,  $\cos \phi_S = \cos 31^\circ 27' = \mathbf{0.853 \text{ lag}}$

(iv) Sending end power =  $3 V_S I_S \cos \phi_S = 3 \times 40,145 \times 100 \times 0.853$

$$= 10273105 \text{ W} = 10273.105 \text{ kW}$$

$$\text{Power delivered} = 10,000 \text{ kW}$$

$\therefore$  Transmission efficiency =  $\frac{10,000}{10273.105} \times 100 = \mathbf{97.34\%}$

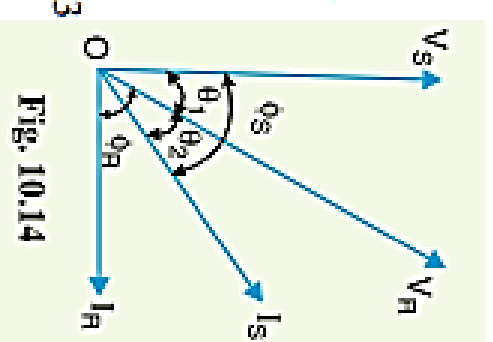
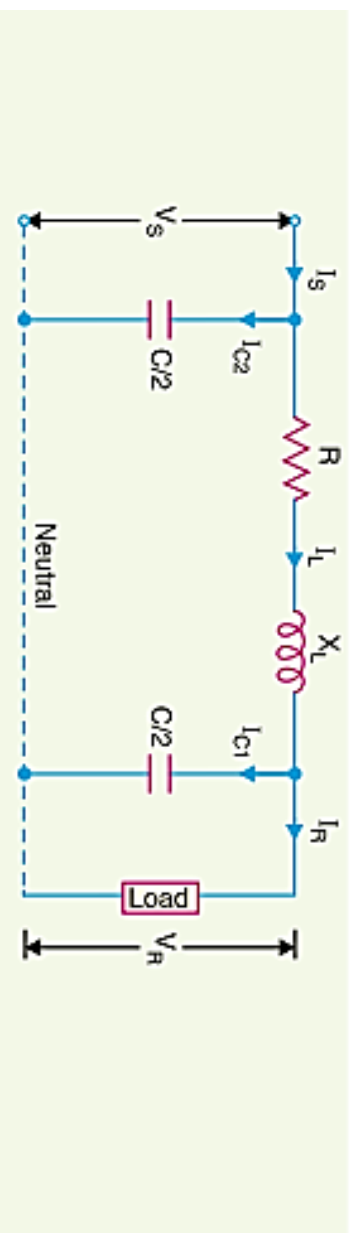


Fig. 10.14

# Nominal $\pi$ method

- In this method, capacitance of each conductor (*i.e.*, line to neutral) is divided into two halves; one half being lumped at the sending end and the other half at the receiving end as shown in Fig.
- It is obvious that capacitance at the sending end has no effect on the line drop.
- However, its charging current must be added to line current in order to obtain the total sending end current.



Let

$I_R$  = load current per phase

$R$  = resistance per phase

$X_L$  = inductive reactance per phase

$C$  = capacitance per phase

$\cos \phi_R$  = receiving end power factor (*lagging*)

$V_S$  = sending end voltage per phase

The \*phasor diagram for the circuit is shown in Fig. 10.17. Taking the receiving end voltage as the reference phasor, we have,

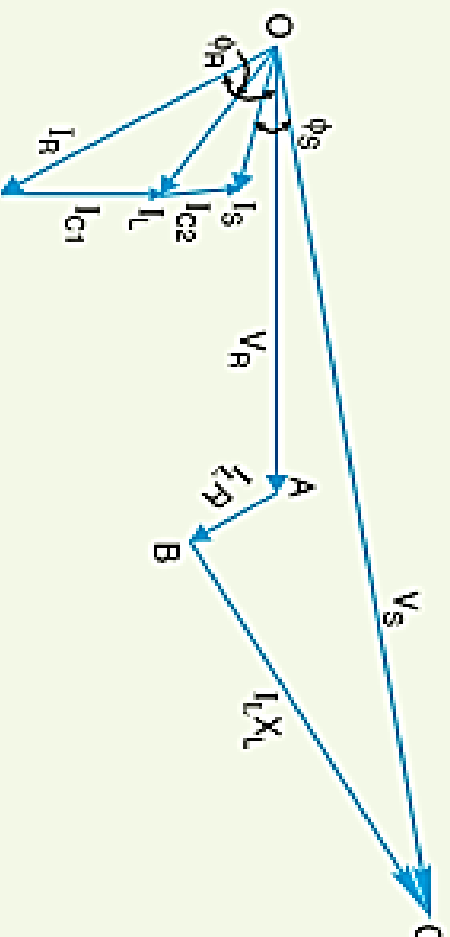
$$\vec{V}_R = V_R + j0$$

Load current,

$$\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R)$$

Charging current at load end is

$$\vec{I}_C = j \omega (C/2) \vec{V}_R = j \pi f C \vec{V}_R$$



# Tutorial 5

*1. A 3-phase, 50Hz, 150 km line has a resistance, inductive reactance and capacitive shunt admittance of  $0.1 \Omega$ ,  $0.5 \Omega$  and  $3 \times 10^{-6} S$  per km per phase. If the line delivers 50 MW at 110 kV and 0.8 p.f. lagging, determine the sending end voltage and current. Assume a nominal  $\pi$  circuit for the line.*

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**Solution.** Fig. 10.18 shows the circuit diagram for the line.

Total resistance/phase,

$$R = 0.1 \times 150 = 15 \, \Omega$$

Total reactance/phase,

$$X_L = 0.5 \times 150 = 75 \, \Omega$$

Capacitive admittance/phase,  $Y = 3 \times 10^{-6} \times 150 = 45 \times 10^{-5} \, \text{S}$

Receiving end voltage/phase,  $V_R = 110 \times 10^3 / \sqrt{3} = 63,508 \, \text{V}$

Load current,

$$I_R = \frac{50 \times 10^6}{\sqrt{3} \times 110 \times 10^3 \times 0.8} = 328 \, \text{A}$$

$$\cos \phi_R = 0.8; \quad \sin \phi_R = 0.6$$

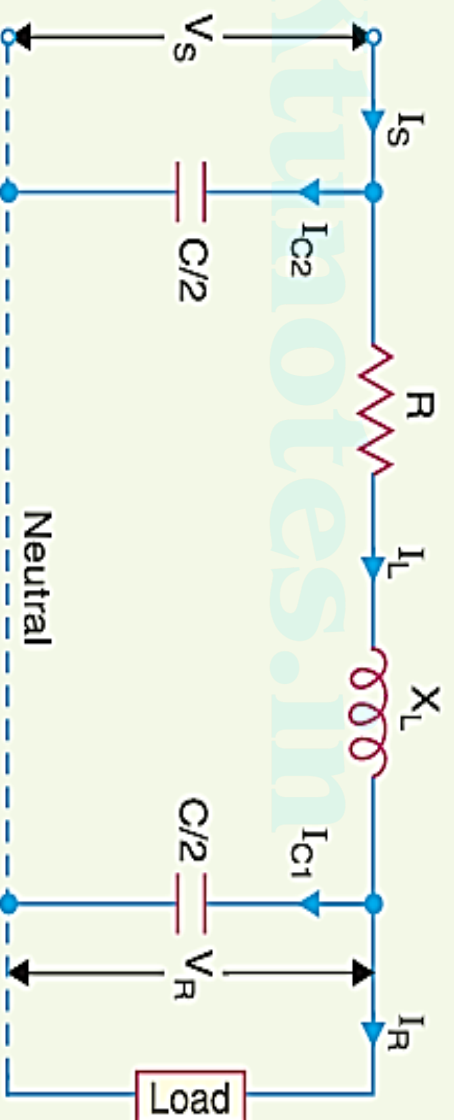


Fig. 10.18

Taking receiving end voltage as the reference phasor, we have,

$$\vec{V}_R = V_R + j0 = 63,508 \, \text{V}$$

Load current,  $\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R) = 328 (0.8 - j0.6) = 262.4 - j196.8$

Charging current at the load end is

$$\vec{I}_{C1} = \vec{V}_R j \frac{Y}{2} = 63,508 \times j \frac{45 \times 10^{-5}}{2} = j14.3$$

Line current,  $\vec{I}_L = \vec{I}_R + \vec{I}_{C1} = (262.4 - j196.8) + j14.3 = 262.4 - j182.5$

Sending end voltage,

$$\begin{aligned} \vec{V}_S &= \vec{V}_R + \vec{I}_L \vec{Z} = \vec{V}_R + \vec{I}_L (R + jX_L) \\ &= 63,508 + (262.4 - j182.5) (15 + j75) \\ &= 63,508 + 3936 + j19,680 - j2737.5 + 13,687 \\ &= 81,131 + j16,942.5 = 82,881 \angle 11.947^\circ \text{ V} \end{aligned}$$

$\therefore$  Line to line sending end voltage =  $82,881 \times \sqrt{3} = 1,43,550 \text{ V} = \mathbf{143.55 \text{ kV}}$

Charging current at the sending end is

$$\begin{aligned} I_{C2} &= j\vec{V}_S Y / 2 = (81,131 + j16,942.5) j \frac{45 \times 10^{-5}}{2} \\ &= -3.81 + j18.25 \end{aligned}$$

Sending end current,

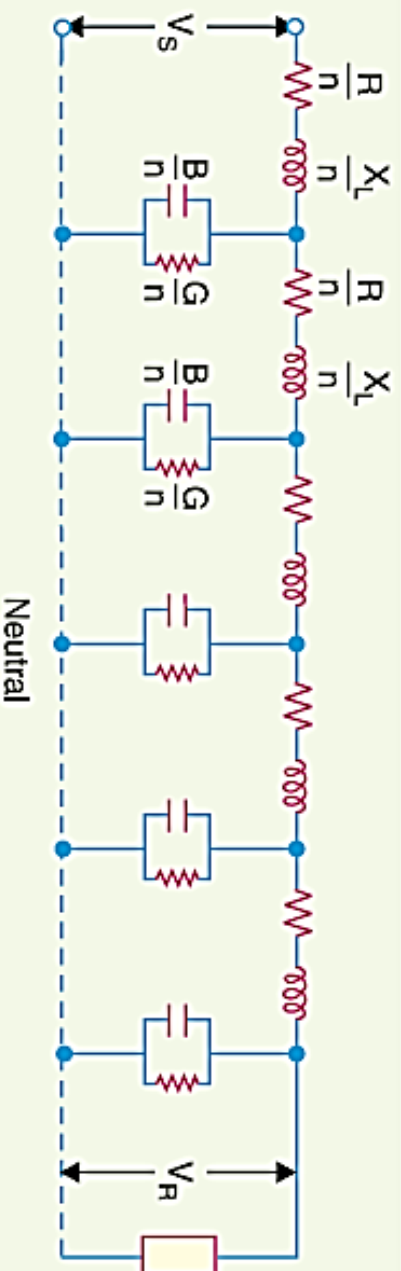
$$\begin{aligned} \vec{I}_S &= \vec{I}_L + \vec{I}_{C2} = (262.4 - j182.5) + (-3.81 + j18.25) \\ &= 258.6 - j164.25 = 306.4 \angle -32.4^\circ \text{ A} \end{aligned}$$

$\therefore$  Sending end current = **306.4 A**



# Long transmission lines

- It is well known that line constants of the transmission line are uniformly distributed over the entire length of the line.
- However, reasonable accuracy can be obtained in line calculations for short and medium lines by considering these constants as lumped.
- If such an assumption of lumped constants is applied to long transmission lines (having length excess of about 150 km), it is found that serious errors are introduced in the performance calculations.
- Therefore, in order to obtain fair degree of accuracy in the performance calculations of long lines, the line constants are considered as uniformly distributed throughout the length of the line.
- Rigorous mathematical treatment is required for the solution of such lines



- Fig shows the equivalent circuit of a 3-phase long transmission line on a phase-neutral basis. The whole line length is divided into  $n$  sections, each section having line constants  $\frac{1}{n}$ th of those for the whole line. The following points may be noted :

- The line constants are uniformly distributed over the entire length of line as is actually the case.
- The resistance and inductive reactance are the series elements.
- The leakage susceptance ( $B$ ) and leakage conductance ( $G$ ) are shunt elements. The leakage susceptance is due to the fact that capacitance exists between line and neutral. The leakage conductance takes into account the energy losses occurring through leakage over the insulators or due to corona effect between conductors. Admittance  $= \sqrt{G^2 + B^2}$ .
- The leakage current through shunt admittance is maximum at the sending end of the line and decreases continuously as the receiving end of the circuit is approached at which point its value is zero.

# Analysis of long Transmission lines (Rigorous solution)

Fig shows one phase and neutral connection of a 3-phase line with impedance and shunt admittance of the line uniformly distributed.

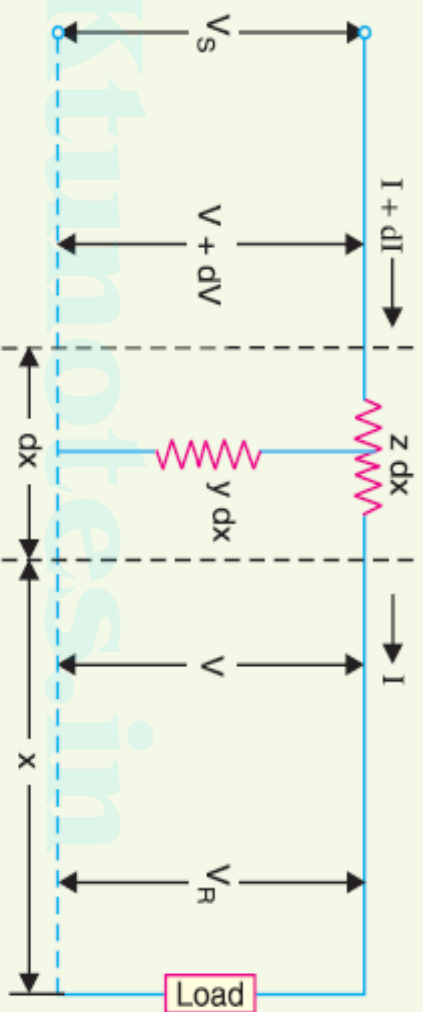


Fig. 10.22

Consider a small element in the line of length  $dx$  situated at a distance  $x$  from the receiving end.

Let  $z$  = series impedance of the line per unit length

$y$  = shunt admittance of the line per unit length

$V$  = voltage at the end of element towards receiving end

$V + dV$  = voltage at the end of element towards sending end

$I + dI$  = current entering the element  $dx$

$I$  = current leaving the element  $dx$

Then for the small element  $dx$ ,

$$z dx = \text{series impedance}$$

Obviously,  $dV = IZ dx$

$$\text{or } \frac{dV}{dx} = IZ \quad \dots(i)$$

Now, the current entering the element is  $I + dI$  whereas the current leaving the element is  $I$ . The difference in the currents flows through shunt admittance of the element i.e.,

$$dI = \text{Current through shunt admittance of element} = VY dx$$

$$\text{or } \frac{dI}{dx} = VY \quad \dots(ii)$$

Differentiating eq. (i) w.r.t.  $x$ , we get,

$$\frac{d^2 V}{dx^2} = Z \frac{dI}{dx} = Z(VY) \quad \left[ \because \frac{dI}{dx} = VY \text{ from exp. (ii)} \right]$$

$$\text{or } \frac{d^2 V}{dx^2} = YZ V \quad \dots(iii)$$

The solution of this differential equation is

$$V = k_1 \cosh(x\sqrt{YZ}) + k_2 \sinh(x\sqrt{YZ}) \quad \dots(iv)$$

Differentiating exp. (iv) w.r.t.  $x$ , we have,

$$\frac{dV}{dx} = k_1 \sqrt{y z} \sinh (x \sqrt{y z}) + k_2 \sqrt{y z} \cosh (x \sqrt{y z})$$

But

$$\frac{dV}{dx} = I z \quad \text{[from exp. (i)]}$$

$\therefore$

$$I z = k_1 \sqrt{y z} \sinh (x \sqrt{y z}) + k_2 \sqrt{z y} \cosh (x \sqrt{y z})$$

or

$$I = \sqrt{\frac{y}{z}} \left[ k_1 \sinh (x \sqrt{y z}) + k_2 \cosh (x \sqrt{y z}) \right] \quad \dots (v)$$

- Equations (iv) and (v) give the expressions for  $V$  and  $I$  in the form of unknown constants  $k_1$  and  $k_2$ . The values of  $k_1$  and  $k_2$  can be found by applying end conditions as under :

At  $x = 0$ ,  $V = V_R$  and  $I = I_R$

Putting these values in eq. (iv), we have,

$$V_R = k_1 \cosh 0 + k_2 \sinh 0 = k_1 + 0$$

$\therefore$

$$V_R = k_1$$

Similarly, putting  $x = 0$ ,  $V = V_R$  and  $I = I_R$  in eq. (v), we have,

$$I_R = \sqrt{\frac{y}{z}} \left[ k_1 \sinh 0 + k_2 \cosh 0 \right] = \sqrt{\frac{y}{z}} \left[ 0 + k_2 \right]$$

$\therefore$

$$k_2 = \sqrt{\frac{z}{y}} I_R$$

Substituting the values of  $k_1$  and  $k_2$  in eqs. (iv) and (v), we get,

$$V = V_R \cosh (x \sqrt{y z}) + \sqrt{\frac{z}{y}} I_R \sinh (x \sqrt{y z})$$

and

$$I = \sqrt{\frac{Y}{Z}} V_R \sinh (x \sqrt{Y Z}) + I_R \cosh (x \sqrt{Y Z})$$

The sending end voltage ( $V_S$ ) and sending end current ( $I_S$ ) are obtained by putting  $x = l$  in the above equations i.e.,

$$V_S = V_R \cosh (l \sqrt{Y Z}) + \sqrt{\frac{Z}{Y}} I_R \sinh (l \sqrt{Y Z})$$

$$I_S = \sqrt{\frac{Y}{Z}} V_R \sinh (l \sqrt{Y Z}) + I_R \cosh (l \sqrt{Y Z})$$

$$\text{Now, } l \sqrt{Y Z} = \sqrt{l Y \cdot l Z} = \sqrt{Y Z}$$

$$\text{and } \sqrt{\frac{Y}{Z}} = \sqrt{\frac{Y l}{Z l}} = \sqrt{\frac{Y}{Z}}$$

where  $Y$  = total shunt admittance of the line  
 $Z$  = total series impedance of the line

Therefore, expressions for  $V_S$  and  $I_S$  become :

$$V_S = V_R \cosh \sqrt{Y Z} + I_R \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z}$$

$$I_S = V_R \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} + I_R \cosh \sqrt{Y Z}$$

It is helpful to expand hyperbolic sine and cosine in terms of their power series.

$$\cosh \sqrt{Y Z} = \left( 1 + \frac{Z Y}{2} + \frac{Z^2 Y^2}{24} + \dots \right)$$

$$\sinh \sqrt{Y Z} = \left( \sqrt{Y Z} + \frac{(Y Z)^{3/2}}{6} + \dots \right)$$

# Generalised circuit constants of transmission lines

## ABCD Parameters

In any four terminal\* network, the input voltage and input current can be expressed in terms of output voltage and output current. Incidentally, a transmission line is a 4-terminal network ; two input terminals where power enters the network and two output terminals where power leaves the network.

Therefore, the input voltage ( $\vec{V}_S$ ) and input current ( $\vec{I}_S$ ) of a 3-phase transmission line can be expressed as :

$$\vec{V}_S = \vec{A} \vec{V}_R + \vec{B} \vec{I}_R$$

$$\vec{I}_S = \vec{C} \vec{V}_R + \vec{D} \vec{I}_R$$

where

$$\vec{V}_S = \text{sending end voltage per phase}$$

$$\vec{I}_S = \text{sending end current}$$

$$\vec{V}_R = \text{receiving end voltage per phase}$$

$$\vec{I}_R = \text{receiving end current}$$

and  $\vec{A}$ ,  $\vec{B}$ ,  $\vec{C}$  and  $\vec{D}$  (generally complex numbers) are the constants known as *generalised circuit constants* of the transmission line. The values of these constants depend upon the particular method adopted for solving a transmission line. Once the values of these constants are known, performance calculations of the line can be easily worked out. The following points may be kept in mind :

- (i) The constants  $\vec{A}$ ,  $\vec{B}$ ,  $\vec{C}$  and  $\vec{D}$  are generally complex numbers.
- (ii) The constants  $\vec{A}$  and  $\vec{D}$  are dimensionless whereas the dimensions of  $\vec{B}$  and  $\vec{C}$  are ohms and siemen respectively.
- (iii) For a given transmission line,
$$\vec{A} = \vec{D}$$
- (iv) For a given transmission line,

$$\vec{A} \vec{D} - \vec{B} \vec{C} = 1$$

We shall establish the correctness of above characteristics of generalised circuit constants in the following discussion.

# Determination of Generalised Constants for Transmission lines ABCD

As stated previously, the sending end voltage ( $\vec{V}_S$ ) and sending end current ( $\vec{I}_S$ ) of a transmission line can be expressed as :

$$\begin{aligned}\vec{V}_S &= \vec{A} \vec{V}_R + \vec{B} \vec{I}_R & \dots (i) \\ \vec{I}_S &= \vec{C} \vec{V}_R + \vec{D} \vec{I}_R & \dots (ii)\end{aligned}$$

We shall now determine the values of these constants for different types of transmission lines.

**(i) Short lines.** In short transmission lines, the effect of line capacitance is neglected. Therefore, the line is considered to have series impedance. Fig. 10.23 shows the circuit of a 3-phase transmission line on a single phase basis.

Here,  $\vec{I}_S = \vec{I}_R$

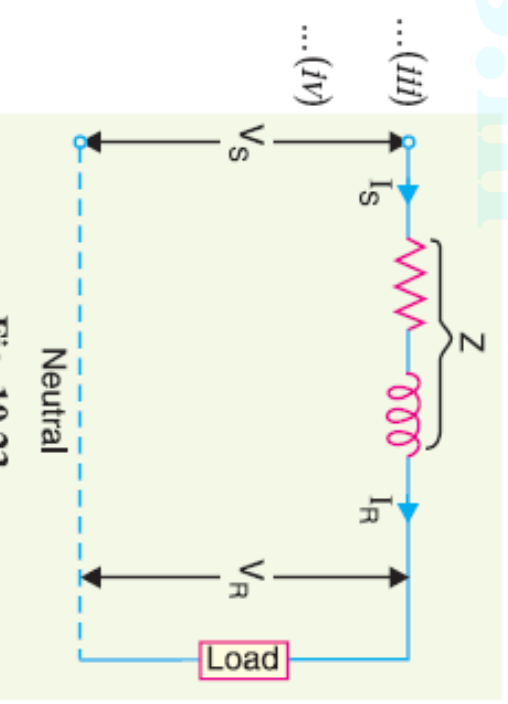
and  $\vec{V}_S = \vec{V}_R + \vec{I}_R \vec{Z}$

Comparing these with eqs. (i) and (ii), we have,

$$\vec{A} = 1 ; \quad \vec{B} = \vec{Z}, \quad \vec{C} = 0 \quad \text{and} \quad \vec{D} = 1$$

Incidentally ;  $\vec{A} = \vec{D}$

and  $\vec{A} \vec{D} - \vec{B} \vec{C} = 1 \times 1 - \vec{Z} \times 0 = 1$

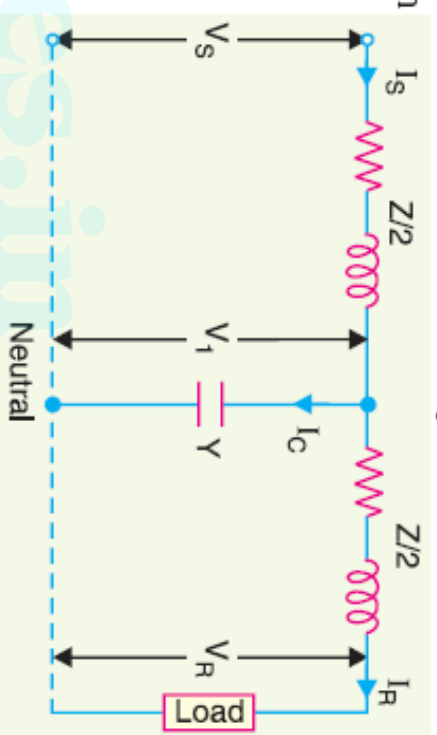


(ii) **Medium lines – Nominal T method.** In this method, the whole line to neutral capacitance is assumed to be concentrated at the middle point of the line and half the line resistance and reactance are lumped on either side as shown in Fig. 10.24.

Here,  $\vec{V}_S = \vec{V}_1 + \vec{I}_S \vec{Z}/2 \quad \dots (v)$

and  $\vec{V}_1 = \vec{V}_R + \vec{I}_R \vec{Z}/2$

Now,  $\vec{I}_C = \vec{I}_S - \vec{I}_R$



$= \vec{V}_1 \vec{Y}$  where  $Y$  = shunt admittance

$$= \vec{Y} \left( \vec{V}_R + \frac{\vec{I}_R \vec{Z}}{2} \right)$$

$$\therefore \vec{I}_S = \vec{I}_R + \vec{Y} \vec{V}_R + \vec{Y} \frac{\vec{I}_R \vec{Z}}{2}$$

$$= \vec{Y} \vec{V}_R + \vec{I}_R \left( 1 + \frac{\vec{Y} \vec{Z}}{2} \right)$$

Substituting the value of  $V_1$  in eq. (v), we get,

$$\vec{I}_S = \vec{I}_R \vec{Z} \cdot \vec{I}_S \vec{Z}$$

Substituting the value of  $I_S$  we get,

$$\vec{V}_S = \left(1 + \frac{\vec{Y}Z}{2}\right) \vec{V}_R + \left(\vec{Z} + \frac{\vec{Y}Z^2}{4}\right) \vec{I}_R$$

Comparing eqs. (vii) and (vi) with those of (i) and (ii), we have,

$$\vec{A} = \vec{D} = 1 + \frac{\vec{Y}Z}{2} ; \quad \vec{B} = \vec{Z} \left(1 + \frac{\vec{Y}Z}{4}\right) ; \quad \vec{C} = \vec{Y}$$

Incidentally :  $\vec{A}\vec{D} - \vec{B}\vec{C} = \left(1 + \frac{YZ}{2}\right)^2 - Z \left(1 + \frac{YZ}{4}\right) Y$

$$= 1 + \frac{Y^2Z^2}{4} + YZ - ZY - \frac{Z^2Y^2}{4} = 1$$

**(iii) Medium lines—Nominal  $\pi$  method.** In this method, line-to-neutral capacitance is divided into two halves ; one half being concentrated at the load end and the other half at the sending end as shown in Fig. 10.25.

Here,  $\vec{Z} = R + jX_L$  = series impedance/phase

$\vec{Y} = j \omega C$  = shunt admittance

$$\vec{I}_S = \vec{I}_L + \vec{I}_{C2}$$

$$\text{or } \vec{I}_S = \vec{I}_L + \vec{V}_S \vec{Y} / 2 \quad \dots(viii)$$

$$\text{Also } \vec{I}_L = \vec{I}_R + \vec{I}_{C1} = \vec{I}_R + \vec{V}_R \vec{Y} / 2 \quad \dots(ix)$$

$$\text{Now } \vec{V}_S = \vec{V}_R + \vec{I}_L \vec{Z} = \vec{V}_R + \left( \vec{I}_R + \vec{V}_R \vec{Y} / 2 \right) \vec{Z} \quad (\text{Putting the value of } \vec{I}_L)$$

$$\therefore \vec{V}_S = \vec{V}_R \left( 1 + \frac{\vec{Y} \vec{Z}}{2} \right) + \vec{I}_R \vec{Z} \quad \dots(x)$$

$$\text{Also } \vec{I}_S = \vec{I}_L + \vec{V}_S \vec{Y} / 2 = \left( \vec{I}_R + \vec{V}_R \vec{Y} / 2 \right) + \vec{V}_S \vec{Y} / 2$$

(Putting the value of  $\vec{I}_L$ )

Putting the value of  $\vec{V}_S$  from eq. (x), we get,

$$\vec{I}_S = \vec{I}_R + \vec{V}_R \frac{\vec{Y}}{2} + \frac{\vec{Y}}{2} \left\{ \vec{V}_R \left( 1 + \frac{\vec{Y} \vec{Z}}{2} \right) + \vec{I}_R \vec{Z} \right\}$$

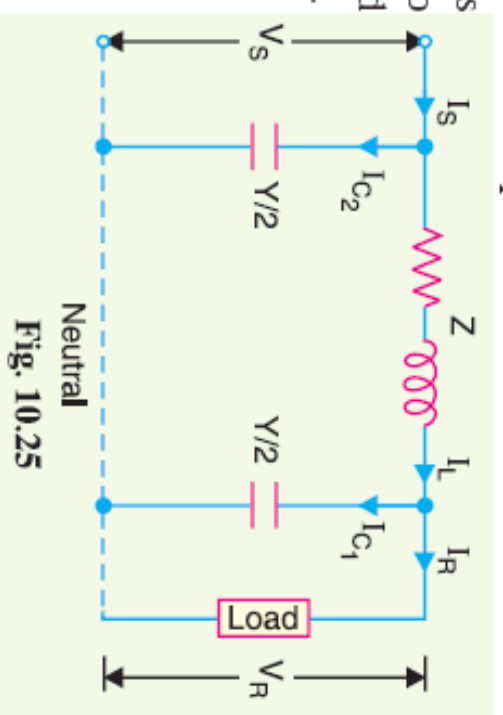


Fig. 10.25

$$\begin{aligned}
&= \vec{I}_R + \vec{V}_R \frac{\vec{Y}}{2} + \frac{\vec{V}_R \vec{Y}}{2} + \frac{\vec{V}_R \vec{Y}^2 \vec{Z}}{4} + \frac{\vec{Y} \vec{I}_R \vec{Z}}{2} \\
&= \vec{I}_R \left( 1 + \frac{\vec{Y} \vec{Z}}{2} \right) + \vec{V}_R \vec{Y} \left( 1 + \frac{\vec{Y} \vec{Z}}{4} \right) \quad \dots (xi)
\end{aligned}$$

Comparing equations (x) and (xi) with those of (j) and (ii), we get,

$$\vec{A} = \vec{D} = \left( 1 + \frac{\vec{Y} \vec{Z}}{2} \right); \quad \vec{B} = \vec{Z}; \quad \vec{C} = \vec{Y} \left( 1 + \frac{\vec{Y} \vec{Z}}{4} \right)$$

Also

$$\begin{aligned}
\vec{A} \vec{D} - \vec{B} \vec{C} &= \left( 1 + \frac{Y Z}{2} \right)^2 - Z Y \left( 1 + \frac{Y Z}{4} \right) \\
&= 1 + \frac{Y^2 Z^2}{4} + Y Z - Z Y - \frac{Z^2 Y^2}{4} = 1
\end{aligned}$$

**(iv) Long lines—Rigorous method.** By rigorous method, the sending end voltage and current of a long transmission line are given by :

$$V_S = V_R \cosh \sqrt{Y Z} + I_R \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z}$$

$$I_S = V_R \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} + I_R \cosh \sqrt{Y Z}$$

Comparing these equations with those of (j) and (ii), we get,

$$\vec{A} = \vec{D} = \cosh \sqrt{Y Z}; \quad \vec{B} = \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z}; \quad \vec{C} = \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z}$$

*Incidentally*

$$\begin{aligned}
\vec{A} \vec{D} - \vec{B} \vec{C} &= \cosh \sqrt{Y Z} \times \cosh \sqrt{Y Z} - \sqrt{\frac{Z}{Y}} \sinh \sqrt{Y Z} \times \sqrt{\frac{Y}{Z}} \sinh \sqrt{Y Z} \\
&= \cosh^2 \sqrt{Y Z} - \sinh^2 \sqrt{Y Z} = 1
\end{aligned}$$

**Example 10.16.** A balanced 3-phase load of 30 MW is supplied at 132 kV, 50 Hz and 0.85 p.f. lagging by means of a transmission line. The series impedance of a single conductor is  $(20 + j52)$  ohms and the total phase-neutral admittance is  $315 \times 10^{-6}$  siemen. Using nominal T method, determine: (i) the A, B, C and D constants of the line (ii) sending end voltage (iii) regulation of the line.

**Solution.** Fig. 10.26 shows the representation of 3-phase line on the single phase basis.

Series line impedance/phase,  $\vec{Z} = (20 + j52) \Omega$

Shunt admittance/phase,  $\vec{Y} = j315 \times 10^{-6} \text{ S}$

(i) **Generalised constants of line.** For nominal T method, various constants have the values as under :

$$\begin{aligned}\vec{A} = \vec{D} &= 1 + \vec{Z} \vec{Y} / 2 = 1 + \frac{20 + j52}{2} \times j315 \times 10^{-6} \\ &= 0.992 + j0.00315 = \mathbf{0.992 \angle 0.18^\circ}\end{aligned}$$

$$\begin{aligned}\vec{B} &= \vec{Z} \left( 1 + \frac{\vec{Z} \vec{Y}}{4} \right) = (20 + j52) \left[ 1 + \frac{(20 + j52)j315 \times 10^{-6}}{4} \right] \\ &= 19.84 + j51.82 = \mathbf{55.5 \angle 69^\circ}\end{aligned}$$

$$\vec{C} = \vec{Y} = \mathbf{0.000315 \angle 90^\circ}$$

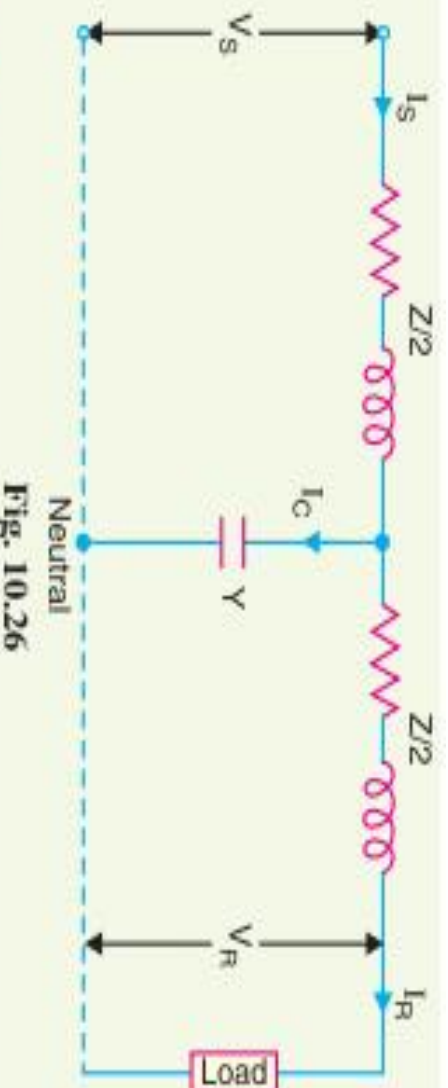


Fig. 10.26

### (ii) Sending end voltage.

Receiving end voltage/phase,  $V_R = 132 \times 10^3 / \sqrt{3} = 76210 \text{ V}$

Receiving end current,  $I_R = \frac{30 \times 10^6}{\sqrt{3} \times 132 \times 10^3 \times 0.85} = 154 \text{ A}$

$\cos \phi_R = 0.85$ ;  $\sin \phi_R = 0.53$

Taking receiving end voltage as the reference phasor, we have,

$$\vec{V}_R = V_R + j0 = 76210 \text{ V}$$

$$\vec{I}_R = I_R (\cos \phi_R - j \sin \phi_R) = 154 (0.85 - j 0.53) = 131 - j 81.62$$

Sending end voltage per phase is

$$\vec{V}_S = \vec{A} \vec{V}_R + \vec{B} \vec{I}_R$$

$$= (0.992 + j 0.0032) 76210 + (19.84 + j 51.82) (131 - j 81.62) \\ = 82,428 + j 5413$$

$\therefore$  Magnitude of sending end voltage is

$$V_S = \sqrt{(82,428)^2 + (5413)^2} = 82.6 \times 10^3 \text{ V} = 82.6 \text{ kV}$$

$\therefore$  Sending end line-to-line voltage

$$= 82.6 \times \sqrt{3} = \mathbf{143 \text{ kV}}$$

**(iii) Regulation.** Regulation is defined as the change in voltage at the receiving end when full-load is thrown off.

Now,

$$\vec{V}_S = \vec{A} \vec{V}_R + \vec{B} \vec{I}_R$$

At no load,

$$\vec{I}_R = 0$$

$\therefore$

$$\vec{V}_S = \vec{A} \vec{V}_{R0}$$

where

$$\vec{V}_{R0} = \text{voltage at receiving end at no load}$$

or

$$\vec{V}_{R0} = \vec{V}_S / \vec{A}$$

or

$$V_{R0} = V_S / A \text{ (in magnitude)}$$

$\therefore$

$$\% \text{ Regulation} = \frac{(V_S / A - V_R)}{V_R} \times 100 = \frac{(82 \cdot 6 / 0 \cdot 992) - 76 \cdot 21}{76 \cdot 21} \times 100 = \mathbf{9 \cdot 25\%}$$

# Ferranti Effect

- A long transmission line draws a substantial quantity of charging current.
- If such a line is open circuited or very lightly loaded at the receiving end the voltage at receiving end may become greater than voltage at sending end.
- This is known as Ferranti Effect and is due to the voltage drop across the line inductance being in phase with the sending end voltages.
- Therefore both capacitance and inductance is responsible to produce this phenomenon.

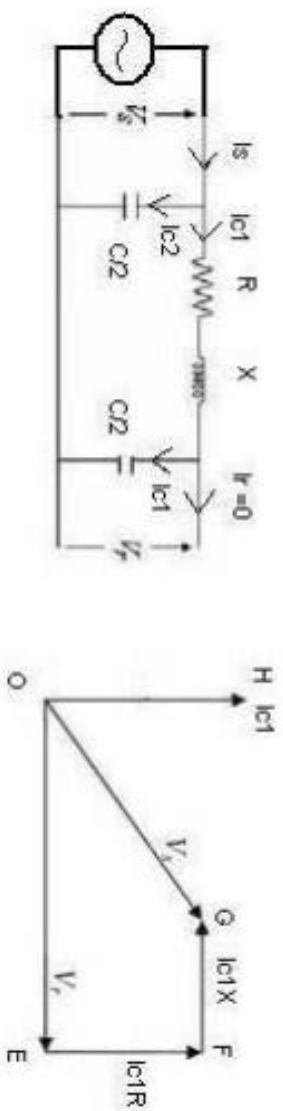


Figure 1. (a) Nominal  $\pi$  model of the line at no load (b) Phasor diagram

- Ferranti Effect can be explained by considering a nominal  $\pi$  model of the line.
- Figure 1(b) shows the phasor diagram of Figure 1(a).
- Here  $OE$  represents the receiving end voltage  $V_r$ .
- $OH$  represents the current  $I_{c1}$  through the capacitor  $C/2$  at the receiving end.
- The voltage drop  $I_{c1}R$  across the resistance  $R$  is shown by  $EF$ .
- It is in phase with  $I_{c1}$ .
- The voltage drop across  $X$  is  $I_{c1}X$ .
- It is represented by the phasor  $FG$  which leads the phasor  $I_{c1}R$  by  $90^\circ$ .
- The phasor  $OG$  represents the sending end voltage  $V_s$  under no-load condition.
- It is seen from the phasor diagram that the receiving end voltage  $V_r$  is greater than the sending end voltage  $V_s$  under no-load condition.

- Ferranti effect can be reduced by installing shunt compensation devices at receiving end. The compensation device is a shunt reactor which is connected in parallel with the transmission line. It reduces the voltage level by absorbing the reactive power.
- Running the transmission line with higher load. i.e if you have two line both carries 5 % load in each side means, we can switch of the one transmission line and the remaining load can be diverted in to another one.

# Proximity Effect

- Before taking this effect we should remember one is electromagnetic induction. i.e all electrical AC motors are running under this principle.
- The current carrying conductor creates magnetic flux, upon the magnetic flux cut by a conductor means the current starts flowing.
- The same effect in the transmission and distribution line is called proximity effect.
- The alternating magnetic flux in a conductor caused by the current flowing in a neighboring conductor gives rise to circulating currents which cause non-uniformity of current and increases the apparent resistance of the conductor.
- This effect is called Proximity effect.

- Proximity Effect is negligible in transmission lines because the distance between the conductor is high. and but in case of cable where the conductors are nearer to each. It is depending upon the size of the conductor, frequency, resistivity and permeability of the material.

### **How Reduce Proximity effect:**

- Reducing the size of the conductor
- Increasing the distance between the two conductor. I.e in cable manufacturing to reduce increase the distance between the conductor, we use dummies.
- Reducing the frequency and increasing the voltage we can reduce the proximity effect. It is practically not possible. However, AC frequency cannot be changed. But we can step up the voltage further steps.